L1. Methodology

Method

What it is?

A method is an empirical practice used in science to produce knowledge. Methods are concrete ways of doing research such as observations experiments or model studies. A method is a tool for collecting evidence or testing claims. A method alone does not explain why it was chosen.

Example(s)

- Observing traffic flow at an intersection
- Running a laboratory experiment on materials
- Using a computer model to simulate weather

Methodology

What it is?

Methodology is the systematic assessment and justification of method choice. It focuses on why one method is chosen instead of another. Methodology depends on the scientific goal and the situation in which the research is done.

Example(s)

- Arguing why a field study is better than a lab experiment for a research question
- Justifying the use of simulations because experiments are not possible

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Conventional view of methodology

What it is?

The conventional view says that scientists choose methods because they are commonly used in their field. The method choice is based on

tradition rather than critical evaluation.

Example(s)

- Always using statistical tests because everyone in the field does
- Following a supervisor's preferred method without questioning it

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Best results view of methodology

What it is?

The best results view says that scientists should choose the method that gives the best results. This view is unclear because it does not explain what best means or how to judge results in advance.

Example(s)

- Choosing a method because it produces results quickly
- Defending a method by claiming it might lead to future benefits

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Epistemic tool view of methodology

What it is?

The epistemic tool view says that methods are tools chosen because we have good reasons to believe they will achieve a scientific goal. Method choice is based on reasons given before the method is used.

Example(s)

- Choosing experiments to find causes
- Using observational studies when experiments are not allowed

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Prediction

What it is?

Prediction is a scientific goal that involves knowing that something will happen in the future. Predictions are often uncertain.

Example(s)

- Predicting the weather
- Predicting how a disease will spread

Explanation

What it is?

Explanation means knowing what caused an event or outcome. An explanation aims to identify causes not just describe patterns.

Example(s)

- Explaining a crash by identifying brake failure
- Explaining climate change by identifying human activity

Design

What it is?

Design is a scientific goal focused on creating objects or systems that perform certain functions. It involves knowing how to make something work as intended.

Example(s)

- Designing a safe bridge
- Creating a medical device

Belief

What it is?

A belief is a mental state where a person thinks that a claim is true. Beliefs can be strong or weak and do not require certainty.

Example(s)

- Believing a theory explains a phenomenon
- Believing a coin toss will land heads

Justification

What it is?

Justification is the evidence or reasons used to defend a belief. In science justification must be possible to explain and defend to others.

Example(s)

- Using data to support a conclusion
- Referring to experiments to defend a claim

Truth

What it is?

Truth means that a claim correctly describes how things really are. A belief can be justified but still false.

Example(s)

- The Earth orbits the Sun
- The geocentric model was once justified but false

Uncertainty

What it is?

Uncertainty means that a belief is not fully certain. Scientific knowledge often includes uncertainty.

Example(s)

- Estimating risk with probabilities
- Making predictions with margins of error

Instrumental reasoning

What it is?

Instrumental reasoning is reasoning about how to achieve a goal. It involves choosing the best means to reach that goal.

Example(s)

- Choosing a method that best fits a research goal
- Selecting simulations when real testing is not possible

Instrumentalism

What it is?

Instrumentalism is the view that a scientific theory is not strictly true or false. A theory is seen as a useful tool for organizing observations and making predictions. The entities the theory talks about are treated as conceptual tools rather than things that actually exist.

Example(s)

- Treating electrons as useful concepts for predicting experimental results without claiming that electrons really exist
- Using a model of ideal gases to make accurate predictions even if the model does not describe reality exactly
- Using economic models that help predict markets without assuming the model reflects how the world truly is

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Realism

What it is?

Realism is the view that a scientific theory can be true or false. According to realism the entities described by a theory either exist or do not exist. Scientific theories aim to describe how the world really is not just to be useful tools.

Example(s)

- Believing that electrons exist as real particles
- Treating gravitational waves as real physical phenomena
- Believing that genes are real entities that cause biological traits

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L2. Scientific Inference

Inference

What is it?

The process of reaching a conclusion from some facts or evidence.

Premise

What is it?

Premise is the starting assumption we take as true for the given problem. it's essentially a statement.

Example:

All balls are green

This object weighs 100kg

The moon is round

Different types of inferences

Direct inference

- Moves from a statement about a sample to a statement about a population.

Examples:

- **Premise**: The sky is blue **and** the grass is green → **Inference**: Therefore, the sky is blue.
- Seeing that

Generalization

What is it?

> We see several cases of something, and we conclude a general rule.

Form:

- We have seen: A, B, C are like this
- So we think: all (or most) things of this kind are like this

Example:

- You see 10 swans, and all of them are white.
- You conclude:
  "All swans in the world are white."

This is not guaranteed to be true (maybe a black swan exists), but it feels reasonable based on what weve seen.

Projection

We have seen a pattern so far, and we **project** it onto a **new case** (often in the future or somewhere else).

Form:

- So far, all Xs we've seen have property Y
- There is a new X we haven't checked yet
- We guess: the new X also has property Y

Example 1 (future):

- Every morning so far, the sun has risen in the east.
- Tomorrow is another morning.
- You conclude:
  "Tomorrow the sun will rise in the east."

Example 2 (medicine):

- A medicine helped many patients with the same illness.
- A new patient has that illness.
- You conclude:
  "This medicine will probably help this new patient too."

Projection is about applying a pattern to a new, unobserved case.

Modus Ponens (deductive)

“If P then Q. P is true. So Q must be true.”

This is a **valid deductive rule**: if the premises are true (i.e premises=statements), the conclusion must be true.

Form:

1. If P, then Q
2. P
3. Therefore, Q

Example:

1. If it is raining, then the streets are wet.
2. It is raining.
3. Therefore, the streets are wet.

If (1) and (2) are true, (3) cannot be false.

Modus tollens(deductive)

If P then Q. Q is not true. So P cannot be true. Like Modus pollens it is also deductive. But here we are doing the inversion. If P→ Q and Q is false then P cannot be true

Form:

1. If P, then Q
2. Not Q
3. Therefore, not P

Example:

1. If the fire alarm is working, then it will sound when there is smoke.
2. There is smoke, but the alarm is not sounding.
3. Therefore, the fire alarm is not working.

Again, if (1) and (2) are true, the conclusion (3) must be true.

Deductive Inferences

What it is:

Deductive inference is a type of reasoning where we start with **general statements or rules**, and from them we derive a conclusion that must be true **if the premises are true**. Deduction does **not add new information**, it simply *unpacks* what is already contained in the premises. Often times its pretty clear from the statement what will follow, its logical.

Example:

1. All humans are mortal.
2. Socrates is a human.
3. Therefore, Socrates is mortal.

Another example:

- All stars burn
- Our sun is a star
- Therefore our sun burns.

There is no way for the conclusion to be false if both premises(statements) are true. That is why deductive reasoning is described as truth-preserving.

The following are Deductive inference:

- Modus Ponens
- Modus Tollens

Inductive Inference

Inductive inference is a type of reasoning where we move from **specific observations** to a **general conclusion**. It goes beyond the information we already have, which means it can be **useful but never guaranteed to be true**. Induction is the kind of reasoning we use in everyday life, science, and prediction. Unlike deductive Inference the conclusion is *probable(might be false)*, not logically certain.

Example:

- You observe that every cat you have seen so far has whiskers.
- You conclude: "All cats have whiskers."

This may be true, but it is not logically guaranteed. One day you might see a cat without whiskers (e.g., due to injury or mutation), which would prove the generalization false. Induction is useful because we cannot observe everything, but we can still make reasonable expectations about the world.

Another example (science):

- Many experiments show that metals expand when heated.
- Scientists conclude: "All metals expand when heated."

  This is not absolutely certain, but strong evidence makes it reasonable to believe.

David Humes critique of inductive inference

Hume asks whether we can justify using induction - that is, whether we have a good reason to trust it. His answer is no, and he gives the following reasoning:

1. We want to justify inductive reasoning

Induction is something we use all the time, for example:

- "The sun has risen every day so far, so it will rise tomorrow."
- "Bread has always fed me, so it will feed me next time I eat it."

1. We cannot justify induction using deduction

   A deductive argument gives a conclusion that must be true if the premises are true.
   But inductive conclusions are not guaranteed - they go beyond what we have observed.
   So we cannot say:
   "The sun has risen every day so far, therefore it must rise tomorrow."
   Because logically, it might not. So deduction cannot save induction.

2. We also cannot justify induction by using more induction

   We might try to defend induction like this:
   "Induction has worked many times in the past, so it will work in the future."

   But this is itself an inductive argument - it assumes the future will be like the past.
   So we would just be using induction to justify induction, which is circular reasoning.

3. Therefore, induction cannot be justified

   Any attempt to defend induction either:

   - fails logically (deduction doesn't work), or

- assumes the very thing it is trying to prove (using induction to justify induction).

So Hume concludes we have no rational proof that induction is reliable, even though we all use it.

Example:

Inductive reasoning:

- Every time I have touched fire, it has burned me.
- So I believe fire will burn me next time too.

Hume's point:

- You cannot deductively prove that the fire will burn you next time - it's not logically guaranteed.
- You also cannot say "It always worked before, so it will work again," because that is already assuming that the future will be like the past - which is exactly what needs to be proven.

So the reasoning is useful, but not logically justified.

Foundationalism

Foundationalism is an argument against David Hume.

What is it?

Foundationalism is the view that knowledge must be built on **basic beliefs** that are certain, secure, and do not need further justification. These basic beliefs act like the foundation of a building: once they are in place, other beliefs can be justified by resting on them. This is used as an argument against David Hume, Foundationalists disagree with Hume because they think there can be secure basic beliefs that do not need justification from past experience.

Example:

Think of the axioms in mathematics, like 0 is a natural number. Or the shortest distance between two points is a straight line. From this we can build theories.

Coherentism

What is it?

They believe that beliefs are not justified by a single foundation, like im happy because I saw my friend. Instead they believe that several beliefs working together and support each other make the belief possible.

Instead of:

> “I am happy because I saw my friend” (just one reason)
>
> Coherentism says:
>
> “I am happy because I saw my friend, I’ve been feeling good lately, the weather is nice, and I’m not stressed.”
>
> Several beliefs together make the main belief more justified

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Example:

Hume says we cannot prove the sun will rise tomorrow, because that is just assuming "the future will be like the past."

A coherentist replies:

We don't believe the sun will rise tomorrow because of one single rule ("it rose before, so it will rise again").
We believe it because this fits with many other beliefs that support each other, such as:

- "The laws of physics"

- "The movement of the Earth"
- "Observations from astronomy"
- "Past experience"
- "Scientific models that successfully predict sunrise times"

Even if one belief failed (e.g., an old astronomy model), the others still hold, so the belief remains justified inside the network.

So Coherentism does not try to escape Hume by proving induction, but instead says Hume's demand for a single foundation is the wrong way to think about justification.

The Hypothetical deductive method (HDM)

1. Formulate a hypothesis H
2. Deduce what observable consequences (C_i) from H
3. Test whether (C_i) is true or not.
4. If (C_i) is false infer that H is false.
5. If (C_i) is true, increase confidence in $`H`$.

Why is HDM both deductive and inductive?

Deductive:

If the hypothesis is true → then we should observe X .
We did not observe X therefore the hypothesis is false.
This is modus tollens, a valid deductive rule.

So deduction is used for falsifying hypotheses.

Inductive:

If the predicted observation does happen, we cannot logically prove the hypothesis is true - we only gain support for it. That is inductive reasoning:

The prediction came true → so the hypothesis is probably right.
But it could still be false in the future, so the support is not certain.

So induction is used for confirming hypotheses.

What is a good hypothesis?

- A statement that is either true or false.
  - Example: All metals expand when heated.

- A statement that is not necessarily true or false.
  - Example : Dark matter interacts only through gravity.

- Has generality or refers to the unobservable
  - Example: Electrons have a negative charge.
  - Example: All dogs need food to survive

What is a good observable sequence $`{(C_i)}`$?

- (C_i) must be observable directly or with the help of accurate measurements (e.g. microscope, X-ray, etc.) Something we can check essentially.

- Deduction must be valid. In other words we should be able to say H is false using {C_i}

- $`{(C_i)}`$must be relevant for H.
  - **Example**:
    A hypothesis H can be that: **Plants need sunlight to survive**, the predicted consequence {C_i} can be : **If the hypothesis is true, then a plant kept in the dark will not grow well.**

Poppers Falsification

What is it?

Karl Popper argued that scientific theories can never be proven true, only proven false. A good scientific theory must make bold, testable predictions - predictions that could turn out to be wrong.

- If a prediction fails, the theory is falsified and must be rejected or revised.
- If a prediction succeeds, the theory is not proven true - it has only survived a test.

So for Popper, science progresses by eliminating false theories, not by confirming true ones.

Key idea:

Science should aim at falsifiability, not verification. A theory is scientific only if it can, in principle, be shown false.

Popper's slogan:
"One counterexample is enough."

Example:

- "All swans are white" is falsifiable - one black swan disproves it.

- "All stars are squared, our sun is round so it disproves it.

Argument against Poppers Falsification

Duhem-Quine Thesis

What is it?

> The idea is that when we test a hypothesis we never a test it alone. Instead we always rely on other assumptions, such as background theories, measurement tools, math, and auxiliary hypotheses (smaller hypothesis = auxiliary hypothesis).

So if a prediction fails, we cannot know for sure which part is wrong:

- the main hypothesis,
- one of the supporting assumptions,
- the experimental setup,
- the data interpretation, etc.

This means that we can't simply say the hypothesis failed because the experiment failed. Instead we need to look into it more carefully and see if we made the wrong assumptions, wrong setup etc.

Example:

> Suppose we test the hypothesis:
> H: "This metal expands when heated."
>
> We heat it and nothing happens. Does that mean H is false?

Not necessarily. Maybe:

- the thermometer was broken,
- the heating device malfunctioned,
- the sample wasn't actually metal,
- the temperature didn't rise enough,
- we used the wrong formula for expansion.

So the failed prediction does not automatically falsify the main hypothesis - something else in the background assumptions might be at fault.

Ad hoc modification

What is it?

Ad hoc modification is when a scientist changes or adds new assumptions to a theory only to protect it from being falsified, instead of improving the theory in a real scientific way.

Example:

Theory:
"All planets move in perfect circles."

Observation:
Mars does not move in a perfect circle.

Ad hoc fix:
"Mars is pushed by an invisible force that we cannot detect."

This saves the theory, but it does not explain anything new, predict anything useful, or make the theory stronger - it only protects it from being proven wrong.

Hypothesis confirmation

Relevance

What is it?

When we test a hypothesis, **not all observations count as good evidence**. A piece of evidence is *relevant* only if it actually makes a difference to whether the hypothesis is likely true or false

Example:

Hypothesis: "This medicine reduces fever."

- Relevant evidence: Patients who take the medicine have lower temperature than those who don't.
- Irrelevant evidence: The pills are blue, the doctor likes the medicine, the patients slept well.

So relevance means: the evidence must be connected to what the hypothesis claims.

Underdetermination

What is it?

Underdetermination means that the same evidence can fit several different hypothesis, so the data by itself cannot tell us which explanation is the right one.

Example:

Data: The population of a city increased every year for 10 years.
Possible hypotheses:

1. "The birth rate is rising."
2. "More people are moving into the city."
3. "Death rates have fallen due to better healthcare."
4. "A combination of all three."

The same data fits multiple explanations - so the hypothesis is underdetermined by the evidence.

L3. Observation and Measurement

Empiricism and Types of Observation

Empiricism

- **Empiricism**: the view that **sensory experience** is the ultimate source of knowledge.
- In the end, scientific claims must connect back to **what we can observe** with our senses, sometimes helped by instruments.

Direct Observation

- You use **only your own senses** to experience the event or object.
- **Examples:**
  - Seeing that a car is moving and not standing still.
  - Feeling that one bag is heavier than another.
  - Hearing that one sound is louder than another.
- **Limitations of direct observation**:
  - It can be **mistaken**:
    - A color blind person can misjudge colors.
    - Strange lighting can change how colors or shapes appear.
  - It is usually **qualitative**, not numerical:

    - You can tell "this is warmer than that", but not exactly *how much* warmer.
    - You can tell "this bag feels heavier", but not its exact weight in kilograms.

Aided Direct Observation

- You still perceive the **thing itself**, but with help from an instrument.
- **Examples**:
    - Glasses or contact lenses to see more sharply.
    - A microscope to see cells or bacteria.
    - A telescope to see distant planets or moons.
- These observations often depend on **background theory**:
    - To trust a telescope, we rely on a theory of **optics** that tells us how lenses bend light.
    - In Galileo's time, some scholars thought telescopes produced illusions, because they did not yet trust the theory.
- **Key idea:**
    - Aided direct observation is still "looking at the object", but the reliability of what you see depends on a theory of how the instrument works.

Indirect Observation

- You do **not** observe the phenomenon itself.

  Instead, you observe an **effect** that the phenomenon causes.
- General structure:

  > Phenomenon (hidden or hard to observe)
  >
  > → causes some effect
  >
  > → we observe the effect and infer the phenomenon
- Examples:
    - We do not see **gravity**, but we see planets move in orbits and objects fall.

    - We do not see **elementary particles**, but we see **tracks** in a bubble chamber.
    - We do not see **temperature itself**, but we see mercury rise in a thermometer.
- **Indirect observations are always theory dependent**:
    - We need a theory that explains how the phenomenon produces the observable effect.
    - Without that theory, the effect would just be "bubbles in liquid" or "lines on a screen" and would not count as evidence for particles or fields.

Theory Dependence of Observation

- An observation is **theory dependent** if its **interpretation** depends on some accepted theory.
- Example with a microscope:
    - A scientist says "I see bacteria".
    - Strictly speaking, they see tiny shapes or dots in a microscope image.
    - It is the **theory of optics** plus **cell theory** and **staining techniques** that justifies calling those dots "bacteria".
- This shows:
    - Many observations, especially aided and indirect ones, are **not neutral**.
    - They already presuppose some theoretical assumptions in order to count as observations *of* something specific.

Logical Empiricism and Circularity

Logical Empiricism

- **Logical empiricism** is a philosophical movement with roughly these ideas:
    - Meaningful statements are either:
        - **empirically testable**, or
        - **logically or mathematically true**.

    - Metaphysical statements that cannot be tested are seen as **meaningless** for science.
    - Philosophy should clarify scientific language and logic, not introduce new untestable claims.
- Examples of how a logical empiricist classify statements:
    - "Electrons have negative charge" → meaningful, because it is testable in principle.
    - "God exists beyond space and time" → not meaningful for science, because it is not testable.
    - "2 + 2 = 4" → meaningful, because it is analytically or logically true.
- Problem highlighted in the chapter:
    - Logical empiricism needs a **clear distinction** between:
        - observational sentences, and
        - theoretical sentences.
    - If many observations are **theory dependent**, this clear separation becomes difficult or impossible.

Threat of Circularity in Indirect Observation

- In indirect observation, we:
    1. Use a **theory** to interpret what we see.
    2. Then use that interpreted observation as **support** for the same theory.
- This risks a kind of **circularity**.

Example from particle physics:

- We see **bubble tracks** in a chamber.
- Theory says:
    - particles cause ionization in the liquid,
    - ionization causes bubbles,
    - the shape of the track is related to charge, mass, and momentum of the particle.
- So we say "We observed a particle with these properties".

- But how do we know the theory about ionization and track shapes is correct?
  - Because it fits the bubble observations.

So:

- The **theory** is used to interpret the observation.
- Then the **interpreted observation** is used to support the theory.
- This is the **threat of circularity** in indirect observation.

The main lesson:

Observation and theory support each other, rather than having a simple one way order where observation comes first and theory comes later.

Operationalization and Operationalism

Operationalization

- **Operationalization** is the process of turning an **abstract or unobservable property** into something we can **measure**.
- Examples of properties of interest:
  - Temperature
  - Stress
  - Intelligence
  - Political attitude
  - Acidity
- Examples of operationalizations:
  - Temperature → height of mercury in a thermometer.
  - Stress → self report questionnaire and cortisol level in saliva.
  - Intelligence → score on an IQ test.
  - Acidity → color of pH paper.
- Basic structure of an operationalization:
  1. **Property of interest**

- What we want to measure, for example temperature or happiness.

2. **Hypothesized causal or correlational link**
   - A theory that says changes in the property lead to changes in some observable effect.
     Example: higher temperature leads to more expansion of a liquid in a thermometer.
3. **Directly observable effect**
   - What we actually look at, for example the height of a liquid column, the color of a strip, or answers on a survey.

A **good** operationalization has:

- A **clear property of interest**.
- A **well justified and sufficiently stable relationship** between the property and the effect.
- An **effect that is easy enough to observe with sufficient precision**.

Operationalism (too strong)

- **Operationalism** is a stronger and more problematic claim:
  - It says that a scientific concept **is nothing more than** the operation or measurement used to detect it.
- Examples of operationalism:
  - "Temperature is what the thermometer reads."
  - "Intelligence is the IQ test score."
  - "Happiness is the number you give on a 0 to 10 scale."
- Problems with operationalism:
  1. **Concept understanding without simple measurement**
     - We can understand concepts even when measurement is very complex or not yet possible, for example extremely small lengths or distances of light years.
  2. **Multiple measurements for the same concept**

    - Different thermometers or sensors can all measure temperature.
      If operationalism is correct, each method would define a different "temperature", which is not how we use the concept in practice.
3. **We want to criticize measurements**
    - In real science, we often ask "Is this test a good measure of intelligence?"
      If "intelligence = IQ score", such criticism makes no sense.

- Conclusion:
    - **Operationalization** is necessary and good.
      It is about making concepts measurable.
    - **Operationalism** is too extreme and is rejected.
      It wrongly identifies the concept with a single measurement method.

Measurement: Steps, Units, and Scales

Steps of Measurement

To measure a property, we typically go through four steps:

1. **Define the concept to be measured**
    - Be clear about what exactly you want to measure.
    - Examples:
        - Temperature = degree of thermal energy of a system.
        - Stress = level of physiological and psychological strain.
        - Intelligence = ability to reason, solve problems, and learn.
2. **Specify the measurement operation**
    - Decide how the property will be detected in practice.
    - Examples:
        - Acidity → use pH paper and see the color change.
        - Stress → use a questionnaire plus cortisol analysis.
        - Length → place a ruler next to the object and see where it ends.
3. **Specify a measure and a unit**

- Choose the instrument and the unit or scale.
- Examples:
  - Temperature → thermometer, unit: Celsius, Fahrenheit, or Kelvin.
  - Weight → scale, unit: kilogram or pound.
  - Time → clock, unit: seconds, minutes.

4. **Represent results with numbers**
   - Record the outcome as a number, for example:
     - 37.2 °C
     - 1.5 kg
     - pH 4
     - stress score 32 out of 40

Requirements on a Good Measure

Two important criteria:

1. **Comparability**
   - The measure must have the **same type of property** as the thing being measured.
   - Examples:
     - A ruler has length. It is comparable to the length of a table.
     - A volume measure (like a measuring cup) has volume. It is comparable to the volume of a liquid.
   - A thermometer cannot directly measure length, because temperature and length are different types of properties.
2. **Stability**
   - The unit must be **sufficiently stable** over time and under normal conditions.
   - Examples:
     - A block of ice is not a stable unit of volume, because it melts and changes shape its not also satisfy comparability as ice and liquid water are not the same.

    - The oscillation of a cesium atom is extremely stable, which is why it is used to define the second.

It is also important that the measurement procedure is **public and repeatable**, so that other people can reproduce the result.

Scale Types

A **scale type** describes what kind of information the numbers carry and which mathematical operations make sense for those numbers.

Another way to define a scale type is by the **transformations of the numbers that leave the meaning of the measurement unchanged**.

Nominal Scale

- **What it is**:
    - Pure **categorization**. The numbers or labels just name different groups.
    - There is no natural order.
- **Examples**:
    - Student ID numbers.
    - Zip codes.
    - Shirt colors (red, blue, green).
    - Blood types, ethnic groups.
- **Allowed transformations:**
    - Any relabeling that preserves **non identity**.
    - For example, if dog = 1 and cat = 2, you can switch to dog = A and cat = B.
      The structure is unchanged as long as different categories stay different.

Ordinal Scale

- **What it is:**
    - A scale that represents **order** or **ranking**.
    - We know which is more or less, but not by how much.

- **Examples**:
  - Race positions: 1st, 2nd, 3rd.
  - TERRIBLE, BAD, NEUTRAL, GOOD, GREAT.
  - Movie ratings with stars.
  - Mohs hardness scale.
- We cannot say that the distance between "BAD" and "NEUTRAL" is equal to the distance between "NEUTRAL" and "GOOD".
- **Allowed transformations:**
  - Any **positive monotone** transformation, which keeps the order.
  - **Example**:

    if ranks are 1, 2, 3, we can transform them to 10, 20, 30 as long as $10 < 20 < 30$.

Interval Scale

- **What it is:**
  - A scale where **equal numerical differences** represent equal differences in the property.
  - Zero is **not a true absence** of the property. It is chosen by convention.
- **Examples:**
  - Temperature in Celsius or Fahrenheit.
  - Calendar years (1000, 1100, 1200).
- We can meaningfully talk about **differences**:
  - 20 °C is 10 degrees warmer than 10 °C.
- We cannot say:
  - 20 °C is twice as hot as 10 °C, because 0 °C is not "no temperature".
- **Allowed transformations:**
  - **Positive linear** transformations:
    $x' = a\,x + b$, with $a > 0$.

- Example: Fahrenheit and Celsius:
  $F = 1.8\ C + 32$.

Ratio Scale

- **What it is:**
  - A scale with equal intervals and a **true zero**.
  - Zero corresponds to a complete absence of the property.
  - Ratios are meaningful.
- **Examples**:
  - Length in meters.
  - Weight in kilograms.
  - Age in years.
  - Temperature in Kelvin.
- **We can say:**
  - 2 meters is twice as long as 1 meter.
  - 80 K is twice the absolute temperature of 40 K.
- **Allowed transformations:**
  - **Positive scalar** transformations:
    $x' = a\ x$, with $a > 0$.
  - Example: meters to inches:
    $\text{inches} = 39.37 \times \text{meters}$.

Absolute Scale

- **What it is:**
  - A scale with a **natural zero** and a **natural unit** that is fixed by the world, not defined by convention.
- **Examples**:
  - Counting objects: 0, 1, 2, 3 apples.
  - Probability: from 0 to 1.

- We cannot rescale these values meaningfully.
  If you have 3 apples, you simply have 3 apples.
- **Allowed transformations:**
  - No non trivial transformations are allowed.
    Any change of the numbers changes the meaning.

Scale Type and Transformations: Summary

- **Nominal**: any transformation that preserves which items are equal or different.
- **Ordinal**: any transformation that preserves the order.
- **Interval**: positive linear transformations (x' = a x + b).
- **Ratio**: positive scalar transformations (x' = a x).
- **Absolute**: no non trivial transformations allowed.

Error, Accuracy, Precision, and Validity

Systematic Error (Bias)

- A **systematic error** is a **consistent, repeatable** error that shifts all measurements in the **same direction**.
- It affects **accuracy**, that is how close the average measurement is to the true value.
- **Examples**:
  - A scale that is miscalibrated and always reads +2 kg too high.
  - A ruler that is slightly too short, so all length measurements are too small.
- A series of measurements can be:
  - **Precise but biased**.
    The results are close to each other but far from the true value.

Random Error (Noise)

- A **random error** causes measurements to **fluctuate unpredictably** around the true value.
- It affects **precision**, that is how tightly repeated measurements cluster together.
- **Example**:
  - You weigh yourself several times and get slightly different numbers each time, even though your real weight is not changing.
- **Important property:**
  - If there is **no systematic error**, the average of many repeated measurements can reduce the impact of random error and get closer to the true value.

Precision vs Accuracy

- **Precision**:
  - Low random error.
  - Repeated measurements cluster tightly.
- **Accuracy**:
  - Low systematic error.
  - Measurements are close to the true value.
- You can have:
  - High precision and low accuracy (clustered but biased).
  - Low precision and reasonable accuracy on average (values scattered around the true value).

Convergent and Divergent Validity

These concepts help us judge whether a measurement method is working properly, and they can also hint at **systematic errors**.

Convergent Validity

- **Convergent validity** is achieved when **different, independent methods** for measuring the **same property** give **similar results**.
- **Example**:
  - Three different thermometers (mercury, gas, infrared) give nearly the same temperature for the same object.
- If they do **not** converge, at least one method may contain **systematic error**.

Divergent Validity

- **Divergent validity** is defined as follows:
  A method has divergent validity if it gives **different results** when applied to **substantially different targets i**f we apply a test on A and another one to B and A and B are **different** then we should observe **different measurements**.
- **Example:**
  - A thermometer should give a low value for and ice cube and a high value for boiling water. If it doesn't, there is an issue because obviously the ice cube is colder.
  - If we measure someones heartbeat when they are running and when they are resting, we should get different results.
- If a measurement method gives almost the **same result for clearly different cases**, it may be insensitive or systematically flawed.

L2. Scientific Inference

Inference

What is it?

The process of reaching a conclusion from some facts or evidence.

Premise

What is it?

Premise is the starting assumption we take as true for the given problem. it's essentially a statement.

Example:

All balls are green

This object weighs 100kg

The moon is round

Different types of inferences

Direct inference

- Moves from a statement about a sample to a statement about a population.

Examples:

- **Premise**: The sky is blue **and** the grass is green → **Inference**: Therefore, the sky is blue.
- Seeing that

Generalization

What is it?

> We see several cases of something, and we conclude a general rule.

Form:

- We have seen: A, B, C are like this
- So we think: all (or most) things of this kind are like this

Example:

- You see 10 swans, and all of them are white.
- You conclude:
  "All swans in the world are white."

This is not guaranteed to be true (maybe a black swan exists), but it feels reasonable based on what weve seen.

Projection

We have seen a pattern so far, and we **project** it onto a **new case** (often in the future or somewhere else).

Form:

- So far, all Xs we've seen have property Y
- There is a new X we haven't checked yet
- We guess: the new X also has property Y

Example 1 (future):

- Every morning so far, the sun has risen in the east.
- Tomorrow is another morning.
- You conclude:
  "Tomorrow the sun will rise in the east."

Example 2 (medicine):

- A medicine helped many patients with the same illness.
- A new patient has that illness.
- You conclude:
  "This medicine will probably help this new patient too."

Projection is about applying a pattern to a new, unobserved case.

Modus Ponens (deductive)

“If P then Q. P is true. So Q must be true.”

This is a **valid deductive rule**: if the premises are true (i.e premises=statements), the conclusion must be true.

Form:

1. If P, then Q
2. P
3. Therefore, Q

Example:

1. If it is raining, then the streets are wet.
2. It is raining.
3. Therefore, the streets are wet.

If (1) and (2) are true, (3) cannot be false.

Modus tollens(deductive)

If P then Q. Q is not true. So P cannot be true. Like Modus pollens it is also deductive. But here we are doing the inversion. If P→ Q and Q is false then P cannot be true

Form:

1. If P, then Q
2. Not Q
3. Therefore, not P

Example:

1. If the fire alarm is working, then it will sound when there is smoke.
2. There is smoke, but the alarm is not sounding.
3. Therefore, the fire alarm is not working.

Again, if (1) and (2) are true, the conclusion (3) must be true.

Deductive Inferences

What it is:

Deductive inference is a type of reasoning where we start with **general statements or rules**, and from them we derive a conclusion that must be true **if the premises are true**. Deduction does **not add new information**, it simply *unpacks* what is already contained in the premises. Often times its pretty clear from the statement what will follow, its logical.

Example:

1. All humans are mortal.
2. Socrates is a human.
3. Therefore, Socrates is mortal.

Another example:

- All stars burn

- Our sun is a star
- Therefore our sun burns.

There is no way for the conclusion to be false if both premises(statements) are true. That is why deductive reasoning is described as truth-preserving.

The following are Deductive inference:

- Modus Ponens
- Modus Tollens

Inductive Inference

Inductive inference is a type of reasoning where we move from **specific observations** to a **general conclusion**. It goes beyond the information we already have, which means it can be **useful but never guaranteed to be true**. Induction is the kind of reasoning we use in everyday life, science, and prediction. Unlike deductive Inference the conclusion is *probable(might be false)*, not logically certain.

Example:

- You observe that every cat you have seen so far has whiskers.
- You conclude: "All cats have whiskers."

This may be true, but it is not logically guaranteed. One day you might see a cat without whiskers (e.g., due to injury or mutation), which would prove the generalization false. Induction is useful because we cannot observe everything, but we can still make reasonable expectations about the world.

Another example (science):

- Many experiments show that metals expand when heated.
- Scientists conclude: "All metals expand when heated."

  This is not absolutely certain, but strong evidence makes it reasonable to believe.

David Humes critique of inductive inference

Hume asks whether we can justify using induction - that is, whether we have a good reason to trust it. His answer is no, and he gives the following reasoning:

1. We want to justify inductive reasoning

Induction is something we use all the time, for example:

- "The sun has risen every day so far, so it will rise tomorrow."
- "Bread has always fed me, so it will feed me next time I eat it."

1. We cannot justify induction using deduction

   A deductive argument gives a conclusion that must be true if the premises are true.
   But inductive conclusions are not guaranteed - they go beyond what we have observed.
   So we cannot say:
   "The sun has risen every day so far, therefore it must rise tomorrow."
   Because logically, it might not. So deduction cannot save induction.

2. We also cannot justify induction by using more induction

   We might try to defend induction like this:
   "Induction has worked many times in the past, so it will work in the future."

   But this is itself an inductive argument - it assumes the future will be like the past.
   So we would just be using induction to justify induction, which is circular reasoning.

3. Therefore, induction cannot be justified

   Any attempt to defend induction either:

   - fails logically (deduction doesn't work), or
   - assumes the very thing it is trying to prove (using induction to justify induction).

   So Hume concludes we have no rational proof that induction is reliable, even though we all use it.

Example:

Inductive reasoning:

- Every time I have touched fire, it has burned me.
- So I believe fire will burn me next time too.

Hume's point:

- You cannot deductively prove that the fire will burn you next time - it's not logically guaranteed.
- You also cannot say "It always worked before, so it will work again," because that is already assuming that the future will be like the past - which is exactly what needs to be proven.

So the reasoning is useful, but not logically justified.

Foundationalism

Foundationalism is an argument against David Hume.

What is it?

Foundationalism is the view that knowledge must be built on **basic beliefs** that are certain, secure, and do not need further justification. These basic beliefs act like the foundation of a building: once they are in place, other beliefs can be justified by resting on them. This is used as an argument against David Hume, Foundationalists disagree with Hume because they

think there can be secure basic beliefs that do not need justification from past experience.

Example:

Think of the axioms in mathematics, like 0 is a natural number. Or the shortest distance between two points is a straight line. From this we can build theories.

Coherentism

What is it?

They believe that beliefs are not justified by a single foundation, like im happy because I saw my friend. Instead they believe that several beliefs working together and support each other make the belief possible.

Instead of:

> “I am happy because I saw my friend” (just one reason)
>
> Coherentism says:
>
> “I am happy because I saw my friend, I’ve been feeling good lately, the weather is nice, and I’m not stressed.”
>
> Several beliefs together make the main belief more justified

---

Example:

Hume says we cannot prove the sun will rise tomorrow, because that is just assuming "the future will be like the past."

A coherentist replies:

We don't believe the sun will rise tomorrow because of one single rule ("it rose before, so it will rise again").
We believe it because this fits with many other beliefs that support each other, such as:

- "The laws of physics"
- "The movement of the Earth"
- "Observations from astronomy"
- "Past experience"
- "Scientific models that successfully predict sunrise times"

Even if one belief failed (e.g., an old astronomy model), the others still hold, so the belief remains justified inside the network.

So Coherentism does not try to escape Hume by proving induction, but instead says Hume's demand for a single foundation is the wrong way to think about justification.

The Hypothetical deductive method (HDM)

1. Formulate a hypothesis H
2. Deduce what observable consequences (C_i) from H
3. Test whether (C_i) is true or not.
4. If (C_i) is false infer that H is false.
5. If (C_i) is true, increase confidence in $`H`$.

Why is HDM both deductive and inductive?

Deductive:

If the hypothesis is true → then we should observe X .
We did not observe X therefore the hypothesis is false.
This is modus tollens, a valid deductive rule.

So deduction is used for falsifying hypotheses.

Inductive:

If the predicted observation does happen, we cannot logically prove the hypothesis is true - we only gain support for it. That is inductive reasoning:

The prediction came true → so the hypothesis is probably right.
But it could still be false in the future, so the support is not certain.

So induction is used for confirming hypotheses.

What is a good hypothesis?

- A statement that is either true or false.
  - Example: All metals expand when heated.

- A statement that is not necessarily true or false.
  - Example : Dark matter interacts only through gravity.

- Has generality or refers to the unobservable
  - Example: Electrons have a negative charge.
  - Example: All dogs need food to survive

What is a good observable sequence $`{(C_i)}`$?

- (C_i) must be observable directly or with the help of accurate measurements (e.g. microscope, X-ray, etc.) Something we can check essentially.

- Deduction must be valid. In other words we should be able to say H is false using {C_i}

- $ `{(C_i)}` $must be relevant for H.
    - **Example**:
      A hypothesis H can be that: **Plants need sunlight to survive**, the predicted consequence {C_i} can be : **If the hypothesis is true, then a plant kept in the dark will not grow well.**

Poppers Falsification

What is it?

Karl Popper argued that scientific theories can never be proven true, only proven false. A good scientific theory must make bold, testable predictions - predictions that could turn out to be wrong.

- If a prediction fails, the theory is falsified and must be rejected or revised.
- If a prediction succeeds, the theory is not proven true - it has only survived a test.

So for Popper, science progresses by eliminating false theories, not by confirming true ones.

Key idea:

Science should aim at falsifiability, not verification. A theory is scientific only if it can, in principle, be shown false.

Popper's slogan:
"One counterexample is enough."

Example:

- "All swans are white" is falsifiable - one black swan disproves it.
- "All stars are squared, our sun is round so it disproves it.

Argument against Poppers Falsification

Duhem-Quine Thesis

What is it?

> The idea is that when we test a hypothesis we never a test it alone. Instead we always rely on other assumptions, such as background theories, measurement tools, math, and auxiliary hypotheses (s maller hypothesis = auxiliary hypothesis).

So if a prediction fails, we cannot know for sure which part is wrong:

- the main hypothesis,
- one of the supporting assumptions,
- the experimental setup,
- the data interpretation, etc.

This means that we can’t simply say the hypothesis failed because the experiment failed. Instead we need to look into it more carefully and see if we made the wrong assumptions, wrong setup etc.

Example:

Suppose we test the hypothesis:
H: "This metal expands when heated."

We heat it and nothing happens. Does that mean H is false?

Not necessarily. Maybe:

- the thermometer was broken,
- the heating device malfunctioned,
- the sample wasn't actually metal,
- the temperature didn't rise enough,
- we used the wrong formula for expansion.

So the failed prediction does not automatically falsify the main hypothesis - something else in the background assumptions might be at fault.

Ad hoc modification

What is it?

Ad hoc modification is when a scientist changes or adds new assumptions to a theory only to protect it from being falsified, instead of improving the theory in a real scientific way.

Example:

Theory:
"All planets move in perfect circles."

Observation:
Mars does not move in a perfect circle.

Ad hoc fix:
"Mars is pushed by an invisible force that we cannot detect."

This saves the theory, but it does not explain anything new, predict anything useful, or make the theory stronger - it only protects it from being proven

wrong.

Hypothesis confirmation

Relevance

What is it?

When we test a hypothesis, **not all observations count as good evidence**. A piece of evidence is *relevant* only if it actually makes a difference to whether the hypothesis is likely true or false

Example:

Hypothesis: "This medicine reduces fever."

- Relevant evidence: Patients who take the medicine have lower temperature than those who don't.
- Irrelevant evidence: The pills are blue, the doctor likes the medicine, the patients slept well.

So relevance means: the evidence must be connected to what the hypothesis claims.

Underdetermination

What is it?

Underdetermination means that the same evidence can fit several different hypothesis, so the data by itself cannot tell us which explanation is the right one.

Example:

Data: The population of a city increased every year for 10 years.
Possible hypotheses:

1. "The birth rate is rising."
2. "More people are moving into the city."
3. "Death rates have fallen due to better healthcare."
4. "A combination of all three."

The same data fits multiple explanations - so the hypothesis is underdetermined by the evidence.

Observation and Measurement

Empiricism and Types of Observation

Empiricism

- **Empiricism**: the view that **sensory experience** is the ultimate source of knowledge.
- In the end, scientific claims must connect back to **what we can observe** with our senses, sometimes helped by instruments.

Direct Observation

- You use **only your own senses** to experience the event or object.
- **Examples:**
  - Seeing that a car is moving and not standing still.
  - Feeling that one bag is heavier than another.
  - Hearing that one sound is louder than another.
- **Limitations of direct observation**:

- It can be **mistaken**:
  - A color blind person can misjudge colors.
  - Strange lighting can change how colors or shapes appear.
- It is usually **qualitative**, not numerical:
  - You can tell "this is warmer than that", but not exactly *how much* warmer.
  - You can tell "this bag feels heavier", but not its exact weight in kilograms.

Aided Direct Observation

- You still perceive the **thing itself**, but with help from an instrument.
- **Examples**:
  - Glasses or contact lenses to see more sharply.
  - A microscope to see cells or bacteria.
  - A telescope to see distant planets or moons.
- These observations often depend on **background theory**:
  - To trust a telescope, we rely on a theory of **optics** that tells us how lenses bend light.
  - In Galileo's time, some scholars thought telescopes produced illusions, because they did not yet trust the theory.
- **Key idea:**
  - Aided direct observation is still "looking at the object", but the reliability of what you see depends on a theory of how the instrument works.

Indirect Observation

- You do **not** observe the phenomenon itself.

  Instead, you observe an **effect** that the phenomenon causes.
- General structure:

> Phenomenon (hidden or hard to observe)
>
> → causes some effect

> → we observe the effect and infer the phenomenon

- Examples:
  - We do not see **gravity**, but we see planets move in orbits and objects fall.
  - We do not see **elementary particles**, but we see **tracks** in a bubble chamber.
  - We do not see **temperature itself**, but we see mercury rise in a thermometer.
- **Indirect observations are always theory dependent**:
  - We need a theory that explains how the phenomenon produces the observable effect.
  - Without that theory, the effect would just be "bubbles in liquid" or "lines on a screen" and would not count as evidence for particles or fields.

Theory Dependence of Observation

- An observation is **theory dependent** if its **interpretation** depends on some accepted theory.
- Example with a microscope:
  - A scientist says "I see bacteria".
  - Strictly speaking, they see tiny shapes or dots in a microscope image.
  - It is the **theory of optics** plus **cell theory** and **staining techniques** that justifies calling those dots "bacteria".
- This shows:
  - Many observations, especially aided and indirect ones, are **not neutral**.
  - They already presuppose some theoretical assumptions in order to count as observations *of* something specific.

Logical Empiricism and Circularity

Logical Empiricism

- **Logical empiricism** is a philosophical movement with roughly these ideas:

    - Meaningful statements are either:
        - **empirically testable**, or
        - **logically or mathematically true**.
    - Metaphysical statements that cannot be tested are seen as **meaningless** for science.
    - Philosophy should clarify scientific language and logic, not introduce new untestable claims.
- Examples of how a logical empiricist classify statements:
    - "Electrons have negative charge" → meaningful, because it is testable in principle.
    - "God exists beyond space and time" → not meaningful for science, because it is not testable.
    - "2 + 2 = 4" → meaningful, because it is analytically or logically true.
- Problem highlighted in the chapter:
    - Logical empiricism needs a **clear distinction** between:
        - observational sentences, and
        - theoretical sentences.
    - If many observations are **theory dependent**, this clear separation becomes difficult or impossible.

Threat of Circularity in Indirect Observation

- In indirect observation, we:
    1. Use a **theory** to interpret what we see.
    2. Then use that interpreted observation as **support** for the same theory.
- This risks a kind of **circularity**.

Example from particle physics:

- We see **bubble tracks** in a chamber.
- Theory says:
    - particles cause ionization in the liquid,

    - ionization causes bubbles,
    - the shape of the track is related to charge, mass, and momentum of the particle.
- So we say "We observed a particle with these properties".
- But how do we know the theory about ionization and track shapes is correct?
    - Because it fits the bubble observations.

So:

- The **theory** is used to interpret the observation.
- Then the **interpreted observation** is used to support the theory.
- This is the **threat of circularity** in indirect observation.

The main lesson:

Observation and theory support each other, rather than having a simple one way order where observation comes first and theory comes later.

Operationalization and Operationalism

Operationalization

- **Operationalization** is the process of turning an **abstract or unobservable property** into something we can **measure**.
- Examples of properties of interest:
    - Temperature
    - Stress
    - Intelligence
    - Political attitude
    - Acidity
- Examples of operationalizations:
    - Temperature → height of mercury in a thermometer.
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L3. Observation and Measurement

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  - Stress
  - Intelligence
  - Political attitude
  - Acidity
- Examples of operationalizations:
  - Temperature → height of mercury in a thermometer.
  - Stress → self report questionnaire and cortisol level in saliva.
  - Intelligence → score on an IQ test.
  - Acidity → color of pH paper.
- Basic structure of an operationalization:
  1. **Property of interest**
     - What we want to measure, for example temperature or happiness.
  2. **Hypothesized causal or correlational link**
     - A theory that says changes in the property lead to changes in some observable effect.

Example: higher temperature leads to more expansion of a liquid in a thermometer.

3. **Directly observable effect**
    - What we actually look at, for example the height of a liquid column, the color of a strip, or answers on a survey.

A **good** operationalization has:

- A **clear property of interest**.
- A **well justified and sufficiently stable relationship** between the property and the effect.
- An **effect that is easy enough to observe with sufficient precision**.

Operationalism (too strong)

- **Operationalism** is a stronger and more problematic claim:
    - It says that a scientific concept **is nothing more than** the operation or measurement used to detect it.
- Examples of operationalism:
    - "Temperature is what the thermometer reads."
    - "Intelligence is the IQ test score."
    - "Happiness is the number you give on a 0 to 10 scale."
- Problems with operationalism:
    1. **Concept understanding without simple measurement**
        - We can understand concepts even when measurement is very complex or not yet possible, for example extremely small lengths or distances of light years.
    2. **Multiple measurements for the same concept**
        - Different thermometers or sensors can all measure temperature. If operationalism is correct, each method would define a different "temperature", which is not how we use the concept in practice.
    3. **We want to criticize measurements**

    - In real science, we often ask "Is this test a good measure of intelligence?"
      If "intelligence = IQ score", such criticism makes no sense.

- Conclusion:
    - **Operationalization** is necessary and good.
      It is about making concepts measurable.
    - **Operationalism** is too extreme and is rejected.
      It wrongly identifies the concept with a single measurement method.

Measurement: Steps, Units, and Scales

Steps of Measurement

To measure a property, we typically go through four steps:

1. **Define the concept to be measured**
    - Be clear about what exactly you want to measure.
    - Examples:
        - Temperature = degree of thermal energy of a system.
        - Stress = level of physiological and psychological strain.
        - Intelligence = ability to reason, solve problems, and learn.
2. **Specify the measurement operation**
    - Decide how the property will be detected in practice.
    - Examples:
        - Acidity → use pH paper and see the color change.
        - Stress → use a questionnaire plus cortisol analysis.
        - Length → place a ruler next to the object and see where it ends.
3. **Specify a measure and a unit**
    - Choose the instrument and the unit or scale.
    - Examples:
        - Temperature → thermometer, unit: Celsius, Fahrenheit, or Kelvin.
        - Weight → scale, unit: kilogram or pound.

    - Time → clock, unit: seconds, minutes.

4. **Represent results with numbers**
    - Record the outcome as a number, for example:
        - 37.2 °C
        - 1.5 kg
        - pH 4
        - stress score 32 out of 40

Requirements on a Good Measure

Two important criteria:

1. **Comparability**
    - The measure must have the **same type of property** as the thing being measured.
    - Examples:
        - A ruler has length. It is comparable to the length of a table.
        - A volume measure (like a measuring cup) has volume. It is comparable to the volume of a liquid.
    - A thermometer cannot directly measure length, because temperature and length are different types of properties.
2. **Stability**
    - The unit must be **sufficiently stable** over time and under normal conditions.
    - Examples:
        - A block of ice is not a stable unit of volume, because it melts and changes shape its not also satisfy comparability as ice and liquid water are not the same.
        - The oscillation of a cesium atom is extremely stable, which is why it is used to define the second.

It is also important that the measurement procedure is **public and repeatable**, so that other people can reproduce the result.

Scale Types

A **scale type** describes what kind of information the numbers carry and which mathematical operations make sense for those numbers.

Another way to define a scale type is by the **transformations of the numbers that leave the meaning of the measurement unchanged**.

Nominal Scale

- **What it is**:
  - Pure **categorization**. The numbers or labels just name different groups.
  - There is no natural order.
- **Examples**:
  - Student ID numbers.
  - Zip codes.
  - Shirt colors (red, blue, green).
  - Blood types, ethnic groups.
- **Allowed transformations:**
  - Any relabeling that preserves **non identity**.
  - For example, if dog = 1 and cat = 2, you can switch to dog = A and cat = B.
    The structure is unchanged as long as different categories stay different.

Ordinal Scale

- **What it is:**
  - A scale that represents **order** or **ranking**.
  - We know which is more or less, but not by how much.
- **Examples**:
  - Race positions: 1st, 2nd, 3rd.
  - TERRIBLE, BAD, NEUTRAL, GOOD, GREAT.
  - Movie ratings with stars.

    - Mohs hardness scale.
- We cannot say that the distance between "BAD" and "NEUTRAL" is equal to the distance between "NEUTRAL" and "GOOD".
- **Allowed transformations:**
    - Any **positive monotone** transformation, which keeps the order.
    - **Example**:

      if ranks are 1, 2, 3, we can transform them to 10, 20, 30 as long as 10 < 20 < 30.

Interval Scale

- **What it is:**
    - A scale where **equal numerical differences** represent equal differences in the property.
    - Zero is **not a true absence** of the property. It is chosen by convention.
- **Examples:**
    - Temperature in Celsius or Fahrenheit.
    - Calendar years (1000, 1100, 1200).
- We can meaningfully talk about **differences**:
    - 20 °C is 10 degrees warmer than 10 °C.
- We cannot say:
    - 20 °C is twice as hot as 10 °C, because 0 °C is not "no temperature".
- **Allowed transformations:**
    - **Positive linear** transformations:
      $x' = a\,x + b$, with $a > 0$.
    - Example: Fahrenheit and Celsius:
      $F = 1.8\,C + 32$.

Ratio Scale

- **What it is:**
    - A scale with equal intervals and a **true zero**.

- Zero corresponds to a complete absence of the property.
- Ratios are meaningful.

- **Examples**:
  - Length in meters.
  - Weight in kilograms.
  - Age in years.
  - Temperature in Kelvin.
- **We can say:**
  - 2 meters is twice as long as 1 meter.
  - 80 K is twice the absolute temperature of 40 K.
- **Allowed transformations:**
  - **Positive scalar** transformations:
    $x' = a\,x$, with $a > 0$.
  - Example: meters to inches:
    inches = 39.37 × meters.

Absolute Scale

- **What it is:**
  - A scale with a **natural zero** and a **natural unit** that is fixed by the world, not defined by convention.
- **Examples**:
  - Counting objects: 0, 1, 2, 3 apples.
  - Probability: from 0 to 1.
- We cannot rescale these values meaningfully.
  If you have 3 apples, you simply have 3 apples.
- **Allowed transformations:**
  - No non trivial transformations are allowed.
    Any change of the numbers changes the meaning.

Scale Type and Transformations: Summary

- **Nominal**: any transformation that preserves which items are equal or different.
- **Ordinal**: any transformation that preserves the order.
- **Interval**: positive linear transformations (x' = a x + b).
- **Ratio**: positive scalar transformations (x' = a x).
- **Absolute**: no non trivial transformations allowed.

Error, Accuracy, Precision, and Validity

Systematic Error (Bias)

- A **systematic error** is a **consistent, repeatable** error that shifts all measurements in the **same direction**.
- It affects **accuracy**, that is how close the average measurement is to the true value.
- **Examples**:
  - A scale that is miscalibrated and always reads +2 kg too high.
  - A ruler that is slightly too short, so all length measurements are too small.
- A series of measurements can be:
  - **Precise but biased**.
    The results are close to each other but far from the true value.

Random Error (Noise)

- A **random error** causes measurements to **fluctuate unpredictably** around the true value.
- It affects **precision**, that is how tightly repeated measurements cluster together.
- **Example**:
  - You weigh yourself several times and get slightly different numbers each time, even though your real weight is not changing.
- **Important property:**

- If there is **no systematic error**, the average of many repeated measurements can reduce the impact of random error and get closer to the true value.

Precision vs Accuracy

- **Precision**:
  - Low random error.
  - Repeated measurements cluster tightly.
- **Accuracy**:
  - Low systematic error.
  - Measurements are close to the true value.
- You can have:
  - High precision and low accuracy (clustered but biased).
  - Low precision and reasonable accuracy on average (values scattered around the true value).

Convergent and Divergent Validity

These concepts help us judge whether a measurement method is working properly, and they can also hint at **systematic errors**.

Convergent Validity

- **Convergent validity** is achieved when **different, independent methods** for measuring the **same property** give **similar results**.
- **Example**:
  - Three different thermometers (mercury, gas, infrared) give nearly the same temperature for the same object.
- If they do **not** converge, at least one method may contain **systematic error**.

Divergent Validity

- **Divergent validity** is defined as follows:
  A method has divergent validity if it gives **different results** when applied to **substantially different targets i**f we apply a test on A and another one to B and A and B are **different** then we should observe **different measurements**.
- **Example:**
  - A thermometer should give a low value for and ice cube and a high value for boiling water. If it doesn't, there is an issue because obviously the ice cube is colder.
  - If we measure someones heartbeat when they are running and when they are resting, we should get different results.
- If a measurement method gives almost the **same result for clearly different cases**, it may be insensitive or systematically flawed.

L4. Experiments

Experiment

Experiments are a *special type* of observational process. What makes them special is that they **actively change the system** in order to learn about causal relationships. An experiment has three defining features: **Control of Background Variables (through manipulation)** , **Intervention on the Target Variable (through manipulation)** **and** **Observation of the Difference Produced by the Intervention.**

Control of Background Variables (through manipulation)

What is it?

A person might have an issue with a machine, before figuring out what caused the failure of the machine (a set of hypothesis) the person must **reduce or eliminate the influence of other factors** that could interfere with the result. The person **sets up conditions** so that only the variable of interest can make a difference.

In simple words:

Remove **anything** that can affect the thing you are trying to test.

Examples:

- Turning off all appliances before testing a single one in the fuse-box example.

- Keeping temperature, humidity, and soil constant when testing fertilizer on plants.
- Giving the control group a placebo so that belief in treatment does not bias results.

Intervention on the Target Variable (through manipulation)

What is it?

Once the background is controlled, the experimenter **changes the independent variable** — the factor believed to cause the effect. This is the **intervention**, and it is most important part. Without intervention, we only have an observational study, not an experiment.

In simple words:

I have made sure that everything that can affect the thing i'm trying to test is removed. Now, I test my hypothesis by changing the thing.

Example:

- Switching on the lamp to see if it blows the fuse.
- Giving a drug to the test group.
- Increasing voltage in an electrical circuit.

Observation of the Difference Produced by the Intervention

What is it?

Now the experimenter **compares the outcome before and after the intervention.** If the results show that there is a difference before and

after the intervention is made, we have a strong case to believe that our hypothesis H_0 is valid.

Example:

- The results show that the fuse blows only when the lamp is turned on → the lamp is the cause.
- The results show that plants grow taller only when fertilizer is used → fertilizer has a positive effect.
- The results show that there weren't any difference in speed after and before we installed new tires on the car → the tires do not improve speed.

Mill's Method of Difference

What is it?

Mill's Method of Difference is a strategy to figure out **what causes a certain effect**.

It goes like this : if you create **two situations that are identical in every way except one factor**, and only one of them shows the effect, then that differing factor is likely the cause.

In simple words:

> If everything is the same except one thing, and only one situation produces the outcome, then that one thing caused it.

- We **control** everything except the suspected cause
- We **intervene** on the suspected cause
- We **observe** whether the effect appears only when the cause is present

  If yes → strong causal evidence

  If no → hypothesis rejected or revised

Examples:

- **Question:** What caused the fuse to blow? (E)
- **Hypothesis:** The lamp caused the fuse to blow (C)
- **S1:** Lamp ON, all other appliances OFF → fuse blows
- **S2:** Lamp OFF, all other appliances OFF → fuse does *not* blow

Since the **only difference** between the two situations was the lamp being turned on, and the fuse blew only in S1, we infer that **the lamp caused the fuse to blow**.

Difference between Observation and Experiment

- **Observation**: We *only watch* what is already happening. No variable is actively changed.

  Goal: Detect patterns, not causes.

  Example: Measuring the temperature of lakes at different depths, or watching the orbits of Saturn's moons.

- **Experiment**: We *actively change* a variable (intervention) and observe the effect while controlling background factors.

  Goal: Test causal hypotheses ("Does X cause Y?").

  Example: Giving one group a new medicine and another a placebo, or changing the pressure in a chamber to see how it affects boiling point

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Observational Study

What it is:

A study where the researcher **does not intervene** in the system but only **observes what is already happening**.

Some background factors may still be *controlled*, but only by choosing when or how to observe not by manipulating the system itself.

In other words:

Just observing. No intervention → no active manipulation of the variable we want to test.

Example:

- Comparing wages of workers in dangerous vs. safe jobs without changing their jobs.
- Recording star movements through a telescope.
- Measuring the temperature of your hand.

Natural Experiment

What it is:

A natural experiment is **not a real experiment**, because we do not **not** change anything.

Instead, something in the real world **happens on its own** and creates two groups that are different in just one important way, almost as if we had set up an experiment.

So we **only observes** what already happened, but the situation is useful for studying cause and effect because the groups can be compared like in a real experiment.

Simpler words:

> Nature (or society) creates the "experiment," not the scientist, i.e we do not make any **intervention** on the variable. The world has already *provided* a situation where the background variables are the same (or similar) except for the one variable of interest.

Why it works:

The background conditions are already controlled by chance, luck, or external events so we can sometimes make causal claims even without doing any intervention.

Example:

- In 1854, John Snow studied a cholera outbreak in London.

  People got water from different wells. One well was contaminated, the others were clean.

  Snow did not assign people to water sources they already lived near them.

  Because the only major difference between the sick and healthy groups was the water source, he could conclude that **dirty water caused cholera**.

Another example:

- A new law is passed in only one city but not in another. Researchers can compare the two cities "as if" one was a treatment group and the other a control group.

Field Experiment

What it is:

A **field experiment** is an experiment done **in the real world** instead of in a laboratory, here we can change the variable of interest (intervention) but we can't successfully control all background variables because it might be difficult.

- We **do intervene** on the independent variable
- But we **cannot fully control** all background variables, because the environment is natural, not artificial.

Example:

- Researcher applies fertilizer to **some** farm plots (intervention)
- Other plots are left unchanged (control group)
- But weather, insects, soil nutrients, etc. **cannot be fully controlled, so** they can affect the results.

Laboratory Experiment

What it is:

A **real experiment** where the researcher not only manipulates the independent variable, but also **controls (or eliminates) nearly all background variables** . Compared to the other experiments like **field experiment** and **natural experiment (note that natural experiment is a observational study** where conditions mimic an experiment**)**, we have full control of the environment and can make interventions.

Example:

- Testing the boiling point of a liquid by changing pressure inside a controlled pressure chamber.
- Chemical reaction performed in a lab under fixed temperature and humidity.
- Medical trials done under controlled hospital/lab conditions with blinding and randomization.

Simulation Experiment

What is it?

A *simulation experiment* is a study where we **build a model (usually on a computer)** that behaves *like* a real experiment, but the things we manipulate are **not real physical variables** —they are **representations** inside the model.

So instead of changing real temperature, real pressure, or real patient behavior, we change **numbers, equations, or programmed conditions** that *stand in* for the real-world variables.

In simpler words:

We manipulate a model of reality, not reality itself.

Therefore, it is *not* a true experiment according to the book chapter 4.

Example:

A city planner simulates traffic flow and tests what happens if a highway exit is removed.

→ Cars in the model are not real cars so the change affects a representation.

Experimental Error

Observer Effect / Influence Problem

What it means:

The act of observing **changes the thing being observed**.

Example:

Putting a thermometer in a liquid slightly changes the liquid's temperature because it must absorb some thermal energy to take a reading .

Or: animals behave differently when they know a human is watching.

Confirmation bias

What is it?:

The results appear distorted because the **observer sees what they expect or want to see**, not what is actually there.

Example:

A researcher who strongly believes the drug works may *unconsciously read* slightly lower pain scores in the treatment group, even when scores are similar.

Placebo Effect

What it means:

A patient improves because they **believe they received treatment**, even if the pill contains no active ingredient.

Example:

A sugar pill reduces pain simply because the patient *expects* it to.

Selection bias

What it means:

The way a sample (or groups within a study) is chosen **is not random**, and ends up being influenced by a factor that affects the outcome.

Example:

Studying the effects of exercise only on people who *chose to join a gym* who are already healthier than average.

Internal Validity

What it means:

A study has **internal validity** when the effect we observe is *really caused* by the factor we are studying and not by some uncontrolled background variable.

> **In simpler words:**
>
> "Did X really cause Y, or was something else responsible?" If this is true, then the change in the dependent variable can be confidently attributed to the intervention, not to chance, bias, or hidden influences.

Example:

A drug trial has high internal validity if the only difference between the treatment and control groups is the drug itself - not age, diet, or stress levels.

Detecting Experimental Errors

Repetition

What it means:

Doing **exactly the same study again**, following the same procedures — same design, same variables, same method.

Example:

Another lab performs the same drug trial using the same protocol and same type of patients.

Reproduction

What it means:

A repetition that **gets the same result** as the original study.

Example:

If the second trial confirms the same drug effectiveness, the result is *reproduced*.

Replication

What it means:

Doing a **modified version** of the study but with slightly different conditions, population, tools, etc. This is to check if the result still holds.

Example:

Testing the same drug but on older patients, or in a different country, or using a different measurement method.

Constancy

What it is:

Keeping background factors **the same** across all groups or trials so they cannot affect the result.

Example:

- Testing two fertilizers on plants but keeping soil type, sunlight, and watering *constant* in both groups.

Elimination

What it is:

Completely **removing** a background factor so it cannot have any influence at all.

Example:

- Dropping objects in a vacuum chamber to eliminate air resistance.
- Using a Faraday cage to eliminate radio interference during measurements.

Binding

What it is:

A strategy to prevent the **observer effect** or **confirmation bias** by **hiding information** from participants or experimenters so they cannot influence the results (consciously or unconsciously).

Single Blinding / Subject Blinding

What it is:

Only the **participants** do not know whether they are in the treatment or control group.

Example:

- In a drug test, patients get either a real pill or a placebo, but they do not know which one.

Experimenter Blinding

What it is:

The **researcher** collecting or interpreting data does **not know** which subjects are in the control or treatment group.

Example:

A lab technician measures response levels but has no information about who received the real drug.

Double Blinding

What it is:

Both **participants** and **experimenters** are blinded, neither knows who is in which group.

Example:

A medical trial where patients do not know if they received a real drug or placebo, and researchers taking measurements also do not know.

This is considered the **gold standard** in clinical experiments.

Separation

What it is

Separation is a strategy used in experiments when we **cannot eliminate** a background factor, but we can still **measure its effect independently** and remove or correct for it afterward.

So instead of:

- **Constancy** → keeping the background factor the same
- **Elimination** → removing the background factor entirely

Separation means:

> "We let the background factor remain, but we monitor it separately so we can mathematically subtract or control for its effect."

In other words, the background factor is **not removed**, but its influence is **isolated and accounted for**.

Example:

- If there is constant ambient sound in the factory, you measure the room noise first, then subtract it from the machine noise.
- If someone is nervous because of the lab environment, you first measure their baseline heart rate before exercise. Then you subtract the baseline from the exercise measurement.

Randomisation in Experiments

Randomization

What is it?

A method where a **random process** (coin flip, lottery, computer RNG) is used to assign units from a sample into **treatment and control groups**.

The goal is to prevent biased assignment and make groups comparable.

Examples:

- Flipping a coin to decide whether each participant gets bright light therapy or placebo.
- Using a random-number generator to divide 100 test samples into two groups.
- Assigning students to two teaching methods by shuffling names digitally.

Random Sampling

What is it?

A method where a **random process** is used to select the sample **from the population**.

This supports **external validity** because the sample is more likely to represent the population.

Examples:

- Drawing 500 names at random from a national database to study mental health.
- Using random-digit dialing to select survey participants.
- Randomly choosing 40 engines from a factory production line for testing.

Data Stratification

What is it?

A procedure where the sample is **divided into subgroups (strata)** based on **known background factors** *before* randomization, so that these factors are balanced across treatment and control groups.

Examples:

- Separating participants by age group (e.g., under 40 / over 40) and randomizing within each group.
- Stratifying patients by disease severity before assigning them to drug vs placebo.
- Grouping plants by initial height before randomizing fertilizer treatment.

External Validity

What is it?

External validity is about how well the results of a study **can be generalized beyond the study itself**.

In other words, it asks:

> “Do the findings hold only for the specific experimental setup, or also in the real world / in other populations / at other times / in other conditions?”

A study has high external validity when its conclusions apply **not just to the sample tested**, but also to the **larger target population or context we care about**.

L.5. Models

Models are purpose dependent

What is it?

Purpose dependence means that a model is only “good” or “bad” **for a specific goal**.

Since all models simplify reality, we cannot judge them in general.

A model is acceptable if its simplifications **do not harm its ability to do the job we need it to do**.

Example:

- **Bohr’s atom model:**

  Not true as a full theory of atoms, but still very helpful for teaching basic atomic structure.

  Good for learning, not good for making precise scientific predictions.

- **Subway map:**

  Shows stations but ignores real distances and street layout.

  Great for planning metro trips, bad for walking around the city.

- **Crash test dummy:**

  Not like a real human in many ways, but good for testing how people might be injured in car crashes.

How is a model different from a theory?

What is the difference?

A **theory** aims to give a *true and complete description* of how some part of the world works.

If a theory is shown to be false, we **reject it**.

A **model**, on the other hand, is allowed to be **false or simplified** as long as it is **useful for a certain purpose**.

We judge a model by how well it works for that purpose, not by whether it describes everything correctly.

In short:

- **Theory:** must be true or accurate in its claims.
- **Model:** may be idealized or simplified—what matters is usefulness.

Examples:

- **Bohr's atom model:**

  As a *theory*, it was rejected when proven false.

  As a *model*, it is still used today because it is useful for teaching.

- **Subway map:**

  Not a true picture of the city, but a very useful model for travel planning.

  A theory could not be this inaccurate.

Positive analogy

What is it?

A positive analogy is a **similarity** between the model and the target.

It is a feature they both share.

Why does it occur in models?

Positive analogies occur because a model must share some important properties with the target in order to represent it and give useful information. Without similarities, the model would tell us nothing about the real system.

Example:

A scale model airplane and a real airplane both have **similar wing shapes**.

A rat and a human both have **similar basic organs and hormones**.

Negative analogy

What is it?

A negative analogy is a **difference** between the model and the target.

It is a feature the model has that the target does not, or the other way around.

Why does it occur in models?

Negative analogies occur because all models **simplify or idealize** reality.

They must differ from the target in some ways, otherwise they would be too expensive, too complex, or impossible to use.

Example:

A scale model airplane is **much smaller** than the real airplane.

Rats are **short-lived** compared to humans.

Neutral analogy

What is it?

A neutral analogy is a property of the model where we **do not yet know** if the target has the same property.

It is an “unknown similarity.”z

Why does it occur in models?

Neutral analogies occur because we often use models to **explore or test new ideas**.

When we manipulate a model, new behaviours appear, and we must check if the real target system behaves the same way.

Example:

A drug works well in rats (model), but we **do not yet know** if it works the same in humans (target).

In Schelling's segregation model, groups cluster but we **don't know** if real cities behave that way without more evidence.

Models vs. Experiments

Similarities:

In both models and experiments, we **set variables and parameters**, we **manipulate** the system, and we **observe** what happens as a result. This means models mimic the basic structure of experimental control, intervention, and measurement.

Differences:

In models, **internal validity** is usually not a big problem because the modeler creates the system and knows exactly how the result is produced. In experiments, internal validity is a major concern because real-world factors can interfere, and we must be sure that the observed effect truly comes from the intervention.

Models also require us to **justify neutral analogies,** we must explain why what happens in the model should also happen in the real target. This is a challenge for all models, but only for some experiments (especially experiments done *not* directly on the real target).

What makes a good model?

Similarity

What is it?

Similarity means that the model shares several important properties with the target.

These shared properties are **positive analogies**, and they must be **relevant for the purpose** of the model.

Examples:

A scale model airplane has **similar wing shape and airflow behaviour** to a real airplane, which makes it useful for studying aerodynamics.

A crash test dummy has **similar body shape and weight distribution** to a human body, which makes it useful for studying crash impacts.

Robustness

What is it?

Robustness means that a model's result stays the same even if you change one of its assumptions.

If the result does not change, that assumption is **not important** for the model's purpose.

Example:

If a mathematical model gives the same outcome whether a parameter is 0.3 or 0.5, the result is **robust** to that parameter.

If changing the exact shape of an airplane wing in a simulation barely affects the predicted lift, the lift result is **robust** to that assumption

Parameter precision

What is it?

Parameter precision means that one model gives a **more detailed or specific** description of its parameters than another.

If the precise model's parameter description **automatically includes** the less precise one, then the first model has higher parameter precision.

Examples:

A model that says the rate of change is **a + b·N** is more precise than a model that only says it is **some function of N**, because a linear function is a specific kind of function.

A model that states a reaction happens at **exactly 25°C** is more precise than one that says it happens at **room temperature**.

Simplicity

What is it?

Simplicity means that a model uses **fewer parameters** or fewer details than another model.

A simpler model is easier to understand and work with, but may be less detailed.

Example:

A model that uses **one parameter** to describe growth is simpler than a model that uses **three parameters**.

A straight-line model is simpler than a curved model because it requires fewer assumptions.

Tractability

What is it?

Tractability means that the model can be **worked with and solved** using reliable methods.

- A model is **computationally tractable** if a computer can calculate its result efficiently.
- A model is **analytically tractable** if you can get the result directly through logical or mathematical deduction.
- A model is **theoretically tractable** if it is either computationally or analytically tractable *and* the steps can be justified using a background theory.

Example(s):

A simple regression model is analytically tractable because you can solve it using algebra.

A simulation of many interacting particles is computationally tractable if the computer can solve it in reasonable time.

A chemical model using Lewis structures is theoretically tractable because it can be solved using functional group theory.

Transparency

What is it?

Transparency means that the user can **understand how the model produces its result**.

If the model's steps are clear enough for a person to follow mentally, it is transparent. If the model is too complex to understand without blindly trusting a computer, it is not transparent.

Example(s):

A simple mathematical model where you can follow each calculation step is transparent.

A large brain simulation with millions of interacting units is not transparent, because no person can fully understand how the final output is produced.

Mirror models

What is it?

Mirror models are models that try to be **as similar as possible** to the real target.

They include many details so the model “mirrors” the real system closely.

Example:

A detailed epidemic simulation that includes real population data, household sizes, workplaces, and movements aims to mirror a real city.

A finite element model of a car body that closely matches the car’s exact shape and material properties is also a mirror model.

Limitations of “models as mirrors”

1. **They become very complex**

   Adding many details makes the model hard to understand, hard to work with, and less transparent.

2. **They lose simplicity and tractability**

   The more details you include, the harder it becomes to calculate results or apply theory.

   You often need heavy computation, and you can’t easily see why a result happens.

3. **They still contain idealizations**

   Even highly detailed models *still* require assumptions, shortcuts, and missing information.

   So they are never truly perfect mirrors.

4. **External validity remains a problem**

Even if the model is very similar to the target, you still need to justify that the assumptions are harmless.

High similarity does **not** remove the need for justification.

5. **They are expensive and time-consuming**

   Collecting huge amounts of real-world data and building detailed models often requires major resources.

6. **Too much data can hide important patterns**

   When everything is included, it becomes harder to identify what really matters causally.

Isolation model

What is it?

An isolation model focuses on **only a few important factors** and ignores the rest.

It is simple and studies one part of a complex system in isolation, instead of trying to mirror everything.

Example:

The Wright brothers tested **lift**, **propulsion**, and **control** separately using simple models before building a full airplane.

A basic economic model that studies supply and demand alone, ignoring all other market details, is also an isolation model.

L6. Statistics

Lying with statistics

What is it?

> Lying with statistics means **using correct statistical tools in the wrong way on purpose**.
>
> The data may be real, and the method may be from a standard statistics book, but the person **chooses the wrong method or wrong presentation** in order to make others believe something false.
>
> The key idea is *intention*: it is only lying if the person **knows** the method is misleading and still uses it.

Example(s):

- Using the **mean** income instead of the **median** to make a family seem richer or poorer, even though the median is the better measure in that situation.
- Showing a graph **without a zero point** on the y-axis to make a small change look big.
- Saying a medical treatment has a **"33% risk reduction"** instead of saying **"1.3 percentage point reduction"**, to make the result look more impressive.
- Reporting that a new medicine "**doubles the risk of blood clots**" without showing that the increase is from **1 in 7000 to 2 in 7000**, which changes how dangerous it really seems.
- Choosing a statistical test that gives the **desired result** instead of the correct one for the situation.

Statistical methodology

What is it?

Statistical methodology is about **explaining why you chose a specific statistical method** for your data and your purpose.

It is not enough to just run a test. You must **justify** why this test is the right one in this situation.

Good methodology helps prevent mistakes and avoids misleading people.

Examples:

- Choosing the **mean** instead of the **median**, and explaining: "I use the mean because I care about total wealth, not inequality."
- Explaining why you use a **t-test** instead of a **chi-square test**, based on the type of data you have.
- Before running a significance test, stating **why** you chose a particular significance level (like 0.05).
- When showing a graph, explaining why you **kept the zero point** to avoid misleading the viewer.
- Choosing **Bayesian methods** because you want to measure *degrees of belief*, and explaining why this fits your goal better than simply rejecting a hypothesis.

Descriptive statistics

What is it?

Descriptive statistics is about **showing data clearly and accurately**.

It uses tools like averages, graphs, tables, and percentages to **summarize** what the data looks like without trying to make deeper predictions.

The goal is to present information in a way that is **honest, clear, and easy to understand**.

Example(s):

- Using the **median** household wealth to show the "middle" person's wealth in a country.

- Making a **bar chart** that starts at zero so the differences are not exaggerated.
- Reporting the **mean temperature** for the year.
- Showing how many people fall into each age group using a **histogram**.
- Using the **absolute risk reduction** (from 3.9% to 2.6%) to clearly communicate how much a medical treatment changes risk.

Inferential statistics

What is it?

Inferential statistics is about **using data to make justified conclusions** about something bigger than the data itself.

You use a sample to make statements about a population, test hypotheses, or estimate effects.

The goal is not just to describe the data, but to **reason beyond it** in a careful and justified way.

Example(s):

- Testing whether a new medicine works by comparing two groups and using a **significance test**.
- Using a **confidence interval** from a sample survey to estimate the true average income of a country.
- Running a **hypothesis test** to see whether a coin is fair.
- Using **Bayesian methods** to update your belief about a hypothesis after seeing new data.
- Determining if a change (e.g., 75% improved after treatment) is **statistically significant** based on sample size and probability.

Stochastic hypothesis

What is it?

A stochastic hypothesis is a hypothesis that **does not give a certain prediction**, but instead says that outcomes happen with **certain probabilities**.

It tells you *how likely* different results are, not which result must happen.

Example(s):

- **Mendel's genetics:**

  When crossing certain pea plants, the hypothesis predicts a **50% chance** of yellow peas and a **50% chance** of green peas—not a guaranteed outcome.

- **Rolling a die:**

  The hypothesis "each side has a 1/6 chance" is stochastic because it states probabilities, not certainties.

- **Weather forecasts:**

  "There is a 40% chance of rain tomorrow" is a stochastic claim.

- **Medical test accuracy:**

  A test might detect a disease with **95% probability**, not with certainty.

Deterministic hypothesis

What is it?

A deterministic hypothesis is a hypothesis that makes **certain**, not probabilistic, predictions.

If the hypothesis is true, then the outcome **must** happen exactly as it says—there is no randomness involved.

Example(s):

- **Chemistry reaction law:**

  “If you heat this substance to 100°C at sea level, it will boil.”

  The prediction is certain given the conditions.

- **Physics (classical mechanics):**

  “If you drop an object in a vacuum, it will fall with constant acceleration.”

  No randomness is involved.

- **Medical example from the lecture:**

  “If a person has Alzheimer’s disease, they will have amyloid plaque in the brain.”

  According to the hypothesis, this must be true for every case.

- **Mathematics:**

  “If a number is even, dividing it by 2 gives an integer.”

  Always true, no exceptions.

Quantitative measure of measurement error

What is it?

A quantitative measure of measurement error is a way to express **how likely it is that an observation is wrong**, using **numbers**.

Instead of simply saying “there might be an error,” we give a **numerical probability** that a mistake in measurement has occurred.

This helps scientists decide **how strongly** an observation should count against a hypothesis.

Example(s):

- A medical test might miss a disease **5% of the time**.

  → The measurement error probability is **0.05**.

- A microscope study might fail to detect amyloid plaque **1 out of 50 times**.

  → Error rate = **0.02**.

- A thermometer may have an accuracy range such that readings are wrong **in 3% of cases**.

  → Error probability = **0.03**.

- In quality control, a machine might incorrectly classify products **2 times per 100 measurements**.

  → Error probability = **0.02**.

Error-based statistics

What is it?

Error-based statistics is about calculating **how probable an observation is if the hypothesis were true**.

Instead of asking "Is the hypothesis true?", we ask:

"If the hypothesis *is* true, how likely is it to see this data?"

If the observation would be **very unlikely** under the hypothesis, then the hypothesis should be **rejected**.

Example(s):

- **Coin fairness test:**

  Hypothesis: "The coin is fair."

  Observation: Only **4 heads in 20 flips**.

  Error-based statistics asks:

  How likely is it to get 4 heads if the coin is fair?

  → If the probability is extremely low, we reject the hypothesis.

- **Genetics experiment:**

  Hypothesis: Offspring traits follow a **50/50** distribution.

  If we observe **90%** of one type, we calculate how probable that is **assuming** the 50/50 model is true.

- **Medical test accuracy:**

  Hypothesis: "This patient does **not** have the disease."

  We ask:

  What is the chance of getting this test result if the patient is disease-free?

  If the chance is very low, we question the hypothesis.

- **Alzheimer's example from the lecture:**

  Hypothesis: "All Alzheimer's patients have plaque."

  If we observe a patient without plaque, we ask:

  How likely is a measurement error?

  If the chance of error is small, the hypothesis is rejected.

Fisher's significance testing

What is it?

Fisher's significance testing is a method for testing a hypothesis by asking:

"If the hypothesis were true, how likely is it to observe data this extreme?"

It uses the **p-value** (a probability) to decide whether the data is too unlikely under the hypothesis.

If the p-value is **smaller** than a chosen significance level (often 0.05), the hypothesis is **rejected**.

The method tests **one hypothesis** at a time and does *not* claim the hypothesis is true—only that it is not rejected.

Example(s):

- **Coin fairness test:**

  Hypothesis: "The coin is fair."

  Observation: 4 heads out of 20 flips.

  Compute p-value = probability of getting 4 or fewer heads if the coin is fair.

  If this probability is very small (like 0.001), we reject the hypothesis.

- **Medicine effect test:**

  Hypothesis: “The medicine has no effect.”

  If the observed improvement would be extremely unlikely under “no effect,” the p-value is low, and the hypothesis is rejected.

- **Classroom experiment:**

  Hypothesis: “Students score equally well before and after training.”

  If most students improve in a way unlikely to happen by chance, the test rejects the hypothesis.

Test statistic

What is it?

A test statistic is **a number you calculate from your sample** that helps decide whether to reject a hypothesis.

It reduces the data to one meaningful value.

Example(s):

- Number of **heads** in 20 coin flips.
- The **average** improvement before/after a treatment.
- The **difference** between two group means.

Sampling distribution

What is it?

A sampling distribution is the **distribution of all possible values** the test statistic can take **if the hypothesis is true**.

It tells you what outcomes are **likely** or **unlikely** under the hypothesis.

Example(s):

- For a fair coin, the sampling distribution gives the probability of getting

0, 1, 2, ..., 20 heads in 20 flips.

---

p-value

What is it?

The p-value is **the probability of getting the observed test statistic, or something even more extreme**, **if the hypothesis is true**.

It measures how surprising your data is under the hypothesis.

Example(s):

- If you get 4 heads in 20 flips, the p-value is the probability of getting **4 or fewer heads** assuming a fair coin.

Significance level

What is it?

The significance level (often **0.05**) is a **threshold** used to decide whether to reject a hypothesis.

If the p-value is **below** this level, the hypothesis is **rejected**.

Example(s):

- p-value = 0.001 < 0.05 → Reject the hypothesis.
- p-value = 0.12 > 0.05 → Do not reject the hypothesis.

---

p-value abuse

What is it?

p-value abuse happens when someone **intentionally changes the way a test is done**—such as the method, the sample, or the way the statistic is calculated—in order to get a p-value that gives the result they *want*.

This is dishonest because the p-value should reflect the actual test, not a manipulated one.

Examples:

- **Repeating the experiment** until one trial gives a “significant” p-value and reporting only that trial.
- **Changing the way the test statistic is counted**, for example grouping outcomes differently to get a lower p-value.
- **Dropping certain data points** that make the result less significant.
- **Switching to a different statistical test** that gives a more favourable p-value.
- **Lowering the significance level** (e.g., from 0.01 to 0.05) to avoid rejecting the hypothesis.

Neyman–Pearson hypothesis testing

What is it?

Neyman–Pearson hypothesis testing is a method for comparing **two opposite hypotheses** and deciding which one to accept.

It focuses on **controlling error rates** (type I and type II errors) and designing a test with enough **power** to make good decisions.

You must always set up **two hypotheses**:

Original hypothesis (Hi)

What is it?

This is the main claim you want to test.

It is treated as the hypothesis that might be rejected.

Example:

Hi: *The coin is fair.*

Alternative hypothesis (Ha)

What is it?

Ha is the **logical opposite** of Hi.

Only one of the two can be true.

If Hi is false, Ha **must** be true.

Example:

Ha: *The coin is not fair.*

Type I error

What is it?

A type I error happens when you **wrongly reject a true Hi**.

Example:

The coin *is* fair, but your test mistakenly concludes it is not.

("False positive.")

Type II error

What is it?

A type II error happens when you **wrongly accept a false Hi**.

Example:

The coin is actually biased, but your test concludes it is fair.

("False negative.")

Power of a test

What is it?

The power of a test is the **probability that the test correctly rejects a false original hypothesis (Hi)**.

In other words, if Hi is actually false, a high-power test will usually detect this and point you toward the alternative hypothesis (Ha).

A test with **low power** often fails to notice real effects.

A test with **high power** is good at detecting when something is wrong with Hi.

Example(s):

- If a medical test has **high power**, it will correctly identify that a disease is present when it really is present.
- In a coin experiment, if the coin is biased but your test only has low power, you might **fail to detect** this bias.
- Increasing the **sample size** (more data) usually increases the power of the test, making results more reliable.

Bayesian statistics

What is it?

Bayesian statistics is a way of evaluating hypotheses by **updating your belief** in a hypothesis after seeing new data.

You start with a **prior probability** (your initial belief), look at the **observed outcome**, and then use **Bayes' theorem** to calculate the **posterior probability**, which is your new, updated belief.

This method treats probability as a **degree of belief**, not just a frequency in data.

Example(s):

- You think there is a **60% chance** a coin is fair (prior).

After flipping it 20 times and getting only 4 heads, you update your belief and maybe conclude there's only a **3% chance** it is fair (posterior).

- Before testing a medicine, you may think it probably works (**prior = 0.7**).

  After seeing strong positive results, your belief increases (**posterior = 0.9**).

- A doctor starts with a low belief a patient has a disease (**prior**).

  After a positive test result, the belief increases according to Bayes' theorem (**posterior**).

The problem of old evidence

What is it?

The problem of old evidence is a difficulty in Bayesian statistics:

It is hard to decide **whether evidence that has already been used to update your beliefs should be used again**.

Bayes' theorem assumes that new data should update your prior beliefs only **once**.

But in real life, people often re-read or re-encounter old evidence, and the theory does not clearly say how to handle this.

Example(s):

- You hear about a scientific experiment and update your belief in a hypothesis.

  Weeks later, you read about the same experiment again and feel like your confidence increases even more — but according to Bayesian rules, you **should not use the same evidence twice**.

- A physicist already believed a theory because of certain experiments ("old evidence").

  When the theory is later formalized, the same old evidence cannot be used again to raise the posterior probability, even though it strongly supports the theory.

- If you tell a friend about your experiment today, and they update their belief,

but then they read your published paper next month and update again,

this “double counting” violates Bayesian principles — but the theory does not clearly say how to prevent it.

The problem of uncertain evidence

What is it?

The problem of uncertain evidence is that Bayesian statistics usually assumes that **all evidence is absolutely correct** (probability = 1).

But in real science, evidence is often **uncertain**, noisy, or possibly wrong.

Classic Bayesian updating does not have a clear way to handle cases where the evidence itself has a probability **less than 1** of being correct.

In other words, Bayes' theorem expects you to update beliefs based on evidence you fully trust but in reality, many observations are **fallible**.

Example(s):

- A medical test might be **95% accurate**, not 100%.

  Bayesian updating says you should update based on “The test is positive,” treating it as certain, even though the test could be wrong.

- A brain scan for amyloid plaque might miss plaque in **some cases**.

  But Bayesianism would treat “Plaque was not observed” as a certain fact, even though it may be an error.

- In physics or biology, measurements can have **random noise** or **equipment error**.

  Standard Bayesian updating does not cleanly incorporate the chance that the measurement is incorrect.

- If a witness reports an event but is only **70% reliable**, Bayesianism does not directly say how to use that partially trustworthy evidence to update beliefs.

The problem of priors

What is it?

The problem of priors is that Bayesianism requires you to **choose a prior probability**—your initial belief in a hypothesis before seeing data—but it does **not give a clear rule** for how this choice should be made.

Different people may choose very different priors, and this can lead to very different posterior probabilities, even with the same data.

The principle principle

What is it?

The principle principle says that if you **know the objective (real-world) probability** of an outcome, then your **prior** should match that probability.

In other words, your belief should follow the actual known chance.

Example:

If a fair die has a 1/6 chance of landing on a 4, your prior for "the next roll is a 4" should be **1/6**, not a random number like 0.3.

The principle of indifference

What is it?

The principle of indifference says that if you **have no information at all** about the chances of different outcomes, then your prior probabilities should be **split equally** among all possibilities.

Example(s):

- If a coin is unfamiliar and you know nothing about it, assign

  P(heads) = 0.5, P(tails) = 0.5

- If a box might contain 3, 4, or 5 balls but you have no clue which, assign

  P(3 balls) = P(4 balls) = P(5 balls) = 1/3

L7. Explanation and Causes

Natural law

What it is?

A **natural law** is a scientific rule that describes a **fundamental and universal pattern in nature**.

It is usually very well-confirmed, applies everywhere, and does not depend on human decisions or culture.

Natural laws are stable, simple, and always true under the same conditions.

Example(s):

- **Newton's law of gravity:** Every mass attracts every other mass.
- **Ohm's law:** Voltage = Current × Resistance.
- **Law of conservation of energy:** Energy cannot be created or destroyed.
- **Boyle's law:** Pressure and volume of a gas are inversely related.

Law-like generalization

What it is?

A **law-like generalization** is a pattern we observe in data that *looks like* a natural law, but is **not guaranteed to be fundamental or universal**.

It is based on empirical observations and might hold most of the time, but there can be exceptions or limits.

Think of it as:

> "This seems to happen regularly in nature, based on what we've seen so far."

Example(s):

- **Bigger animals tend to live longer than smaller ones.** (Often true, but not always.)
- **High-pressure weather systems usually bring clear skies.**
- **In chromatography, larger molecules come out of the column first.**
- **People who exercise regularly tend to live longer.** (A strong pattern, but not a universal law.)

Deductive-Nomological account of explanation

What it is?

It is an idea that says we explain something by **using a scientific law plus facts about the situation** to **logically show why the event had to happen**.

In other words:

Law + starting facts = explanation

Example(s):

- **Why did the ball fall?**

  Law: Gravity pulls objects down.

  Fact: You dropped the ball.

  Conclusion: The ball must fall.

  This is a DN explanation.
- **Why is the shadow a certain length?**

  Law: Light creates shadows based on angles.

  Fact: The flagpole is a certain height.

  Conclusion: The shadow must have this length.

Why something can satisfy DN conditions but still not be a real explanation

Reason:

DN only checks if the event can be *logically deduced* from a law and some facts.

It does **not** check whether the explanation points to the **real cause**.

So DN accepts explanations that work on paper but make no sense in reality.

Examples

1. Flagpole and shadow

You can deduce the pole height from the shadow.

DN says it is a valid explanation.

But the shadow does **not** cause the pole's height.

So it is not a real explanation.

---

2. Mr. Jones cannot get pregnant

DN uses the rule "men who take birth-control pills don't get pregnant."

It is logically correct, but the pills are irrelevant.

The real cause is being male.

---

3. Pendulum period and length

DN says either variable can explain the other.

But only the **length causes the period**, not the other way around.

Explanandum

What it is?

The **explanandum** is the thing you want to explain.

It is the **question or event** you are asking about.

Example(s):

- A vase broke.

  → This is the explanandum.
- Why did the car crash?

  → The car crash is the explanandum.
- Why is the sky blue?

  → The blueness of the sky is the explanandum.

Explanans (explanantia)

What it is?

The **explanans** is the information that **gives the explanation**.

It is what you use to explain the thing you are asking about.

Example(s):

- Explanandum: The vase broke

  Explanans: Someone bumped the table and the vase fell.
- Explanandum: The car overheated

  Explanans: The radiator was empty and the engine got too hot.
- Explanandum: The sky is blue

  Explanans: Light scatters in the atmosphere.

Singular causal explanation

What it is?

A **singular causal explanation** explains one specific event by listing the **specific things that caused it**.

It does not use general laws. It uses **particular events** that happened in this case.

Example(s):

- Why did the vase break?

  Because

  1. It was on the edge of the table
  2. You bumped the table
  3. The vase fell and hit the floor

     → These specific events caused the break.
- Why did the car stop?

  Because

  1. The fuel tank was empty
  2. The engine could not get fuel

     → These particular events explain this one situation.

Singular explanandum

What it is?

A **singular explanandum** is one specific event or thing that you want to explain.

Example(s):

- Why did **this vase** break?
- Why did **this car** crash?
- Why is **this person** sick?

General explanandum

What it is?

A **general explanandum** is a general pattern or trend you want to explain.

It is not about one event but about many cases.

Example(s):

- Why do **penguins** live in cold places?
- Why does **inflation** rise when interest rates drop?
- Why do **heavier cars** use more fuel?

Singular explanans

What it is?

A **singular explanans** explains something using **specific events** that happened in this particular case.

Example(s):

- The vase broke because
  1. You hit the table
  2. The vase fell
  3. It hit the floor
- The car stopped because
  1. The fuel tank was empty
  2. The engine lost power

General explanans

What it is?

A **general explanans** explains something by using a **general rule or law**, not specific events.

Example(s):

- Objects fall because of **gravity**.
- Gases expand when heated because of the **gas laws**.
- Larger animals live longer because of **biological scaling laws**.

Contrastive explanandum

What it is?

A **contrastive explanandum** is a question that asks

"Why this instead of that?"

It means you are not just asking why something happened, but why it happened **in this way rather than another possible way**.

Example(s):

- Why did **this car** crash **here** instead of **somewhere else**?
- Why did the vase break **now** instead of **later**?
- Why did the storm hit **Stockholm** instead of **Uppsala**?
- Why did this person choose **A** instead of **B**?

Contrastive explanans

What it is?

A **contrastive explanans** gives the explanation that answers a **contrastive question**.

It explains **why one outcome happened instead of another** by pointing to the **specific difference-making cause**.

Example(s):

- Why did the car crash **here** instead of **down the road**?

  Contrastive explanans: This curve was **tighter**, while the other part of the road was **straight**.

- Why did the storm hit **Stockholm** instead of **Uppsala**?

  Contrastive explanans: The **pressure system** moved east, not north.

- Why did this student pass **instead of** failing?

  Contrastive explanans: They **studied extra** the last week, which made the difference.

Accuracy

What it is?

Accuracy means the explanation must describe the **real world correctly**.

The explanans must mention the **actual cause** that truly made the difference.

It should not include false information or irrelevant causes.

In short:

A good explanation must point to the real difference-making cause.

Example(s):

- The vase broke because the **cat pushed it off the table**.

  (Accurate, because this is what really happened.)

- The vase broke because of **a ghost**.

(Not accurate, because it does not describe the actual cause.)

- The car skidded because the **road was icy**, not because of the color of the car.

  (The icy road is the real difference-making cause; car color is irrelevant.)

Precision (of the explanandum)

What it is?

Precision means the question you are trying to explain must be **clear and specific**.

You must clearly state **what the contrast is**, meaning:

What exactly are we comparing?

Why this instead of what?

A precise explanandum makes it easier to give a good explanation.

Example(s):

- Vague:

  "Why did the car crash?"

  (Not precise, many possible answers.)

- Precise:

  "Why did the car crash **here** instead of **around the corner**?"

  (Clear contrast.)

- Vague:

  "Why did the vase break?"

- Precise:

  "Why did the vase break **today** instead of **yesterday**?"

Difference-Making (of the explanans)

What it is?

The explanans must point to the **specific cause(s)** that actually made the difference in the situation.

It should answer the contrast directly:

Why this outcome instead of the other one?

Only the **difference-making causes** matter.

Extra details that did not affect the outcome should not be included.

Example(s):

- Explanandum: Why did the car crash **here** instead of **further down the road**?

  Difference-making explanans:

  "The curve **here** was too sharp."

  (This is the cause that made the difference.)
- Explanandum: Why did this plant grow **faster** than the other plant?

  Explanans: The faster-growing plant got **more sunlight**.

  (This is the difference-making factor.)
- Explanandum: Why did student A pass **instead of** failing?

  Explanans: They **studied more** during the last week.

  (This directly produced the difference.)

Non-sensitivity (of the explanans)

What it is?

Non-sensitivity means the explanation should **not depend on a long chain of fragile background conditions**.

A good explanans works **even if small details in the background change**.

If the explanation depends on many things all being exactly right, then it is **too sensitive** and not reliable.

In short:

A good explanation should not fall apart if one tiny background detail changes.

Example(s):

- Bad (too sensitive):

  "The revolution happened because Lenin skipped listening to Beethoven that day."

  (This depends on many unrealistic background assumptions.)

- Good (non-sensitive):

  "The revolution happened because of political unrest and economic collapse."

  (These causes do not depend on fragile background conditions.)

Cognitive salience

What it is?

Cognitive salience means the explanation should be **easy to understand** for the person who is asking the question.

A good explanation is clear, simple, and uses ideas the audience can easily grasp.

In short:

The easier the explanation is to understand, the better it is.

Example(s):

- Easy to grasp (good):

  "The car skidded because the road was icy."

  (Simple and clear.)

- Hard to grasp (not good):

  “The car skidded due to complex friction coefficients and micro-level tire interactions.”

  (True but too complicated for most people.)

- Easy to grasp (good):

  “The plant died because it got no water.”

- Hard to grasp (not good):

  “The plant died because of a drop in cellular turgor pressure caused by osmotic imbalance.”

  (Scientifically correct but not cognitively simple.)

Direct cause

What it is?

X is a **direct cause** of Y if changing X would **immediately change Y**, while everything else stays the same.

In other words:

If you can **intervene on X** and Y changes because of that, then X directly causes Y.

Example(s):

- Turning the **light switch** (X) directly causes the **light to turn on** (Y).

  If you flip the switch, the light changes, even if everything else stays the same.

- Pressing the **brake pedal** (X) directly causes the **car to slow down** (Y).

  Changing X changes Y.

- Heating **water** (X) directly causes the **temperature** (Y) to rise.

  If you add heat, the water gets hotter.

Contributing cause

What it is?

A **contributing cause** is something that helps bring about an outcome **through a chain of direct causes**.

X is a contributing cause of Y if:

- X starts a **causal chain**,
- each step in the chain is a **direct cause**, and
- the chain eventually ends at Y.

X might not directly cause Y, but it still **contributes** to Y happening.

Example(s):

- You **push a domino** (X).

  That domino hits another, and eventually the last one **falls** (Y).

  → X is a contributing cause of Y through the chain.
- You **light a match** (X).

  The flame lights a candle, which heats wax, which melts it (Y).

  → Your action contributed to Y through multiple steps.
- A **traffic jam** (X) slows cars behind it, which delays people, which causes someone to **arrive late** (Y).

  → X contributed to Y through a series of direct effects.

Correlation

What it is?

Correlation means that two variables vary together in a systematic way. When one variable changes the other tends to change as well. Correlation is symmetric. If X is correlated with Y then Y is also correlated with X. Correlation by itself does not show that one variable causes the other.

Example(s)

- Ice cream sales and temperature are correlated
- Shoe size and reading ability are correlated in children because age affects both
- Exercise levels and health outcomes are correlated
- Screen time and sleep quality are correlated

L8 Qualitive methods

Social Science

What it is?

Social science is the study of **human behaviour**, how people think, feel, act, and interact.

It looks at **societies**, **groups**, and **social processes**.

It often uses **interviews, observations, and qualitative data** to understand why people do what they do.

Example(s)

- Studying why people buy certain products
- Observing how students behave in a classroom
- Interviewing people about their experiences
- Research on cultures, politics, or families

Natural Science

What it is?

Natural science is the study of the **physical world**, such as nature, matter, energy, and living organisms.

It uses **experiments, measurements, and quantitative data** to find general laws and patterns in nature.

Example(s)

- Studying how electricity works
- Measuring the speed of falling objects
- Observing how plants grow under different conditions
- Research in physics, chemistry, or biology

Simulation theory of interpretation

What it is?

Simulation theory says that we understand other people by using our own mind as a model.

We imagine what we would think or feel in their situation, and this helps us guess their mental states.

Example(s)

- You see someone looking sad and imagine how you would feel in that moment, so you conclude they are sad.
- You watch someone struggle with a task and mentally "try it yourself" to understand what they might be thinking.

Theory theory of interpretation

What it is?

Theory theory says that we understand other people by forming a kind of mental theory about how minds work.

We collect evidence about their behaviour, make hypotheses about their beliefs or desires, and update these hypotheses when we get new

information.

Example(s)

- You see someone avoid a dog and infer they must believe dogs are dangerous.
- You watch how a friend reacts in many situations and build a general idea of what they usually want or believe, like a small scientific theory about their mind.

Belief desire explanation

What it is?

A belief–desire explanation says that an agent chooses **action A** instead of **action B** because she **desires outcome X** (and not **Y**) and **believes** that **A will lead to X**, while **B will lead to Y**.

Example(s)

- The agent chooses **A** because she wants **X**, and she believes **A → X** and **B → Y**.
- If the agent desires **X** and believes that **only A** brings about **X**, she will pick **A** over **B**.

What are mental states?

Introspective psychology

What it says?

Mental states are **inner experiences** such as thoughts, feelings, desires, and emotions.

We can know them by **looking inside our mind** through introspection.

Example(s)

- "I feel sad."
- "I want to win."
- "I believe it will rain."

Methodological Behaviorism

What it says?

Mental states should **not** be studied because they cannot be observed directly.

Science should only study **behaviour** (what people do), not what they think or feel.

Example(s)

- Instead of saying "She is angry," a behaviourist says:

  "She is shouting and walking fast," because these are observable.

Cognitive science

What it says?

Mental states **do exist**, like beliefs, desires, plans, and intentions.

But we should study them scientifically by connecting them to **observable behaviour** and using evidence not introspection.

Example(s)

- Inferring someone's belief from their choices.
- Using reaction times or errors in tasks to understand mental processes.

The intentional stance

What it is?

The intentional stance is a way of understanding and predicting an agent's behaviour by **treating the agent as if it has beliefs and desires**.

This method can be used for **humans, animals, or even machines**.

We assume the agent is **rational**, meaning it will try to achieve its goals based on what it believes.

Example(s)

- Predicting that a person will take an umbrella because they **believe** it will rain and **desire** to stay dry.
- Saying a dog goes to the kitchen because it **believes** food is there and **wants** to eat.
- Saying a chess computer "wants to protect its queen" because this helps us predict its next move, even though the computer does not literally have desires.

Qualitative data

What it is?

Qualitative data is **data without numerical values**.

It usually involves **descriptions, categories, labels, or classifications** rather than measurements or numbers.

It tells us *what something is like*, not *how much* of it there is.

Example(s)

- Interview answers describing a person's experience.
- Categories such as "happy," "neutral," or "sad."
- Types of user complaints listed as "navigation problem," "slow system," or "confusing layout."
- Photos or videos used to understand behaviour.

Mixed method (of data types)

What it is?

A mixed method uses **both qualitative data and quantitative data** in the same study.

This means researchers use qualitative data **as it is** (for deep understanding) and also **turn some of it into numbers** so it can be measured or compared.

It combines the strengths of both data types.

Example(s)

- Interview texts are read for meaning (qualitative), and then certain words or themes are counted (quantitative).
- A focus group discussion is analyzed for emotions and attitudes (qualitative), and then researchers code each emotion into categories and measure how often each appears (quantitative).
- Photos from observations are described (qualitative) and then classified into groups for statistical analysis (quantitative).

Interpretation

Main problem of interpretation

What it is?

The main problem of interpretation is that **many different mental-state explanations can fit the same behaviour**.

A single action can be explained by several **plausible beliefs and desires**, and these are called **alternative mental-state models**.

Because of this, interpretation is not always reliable.

How this problem can be faced

1. **Avoiding interpretation**

   Use methods that depend less on guessing mental states, such as studying behaviour, social patterns, or physiological signals.

2. **Improving interpretation**

   Gather more evidence, check for consistency, repeat questions, use multiple observers, and apply clear quality criteria to make interpretations more trustworthy.

Example(s)

- If someone rejects an offer, they might be angry, confused, proud, or following a social rule. Many explanations fit the same action.
- Observing more behaviour, asking questions, or checking with other observers helps decide which mental-state model is most likely.

Case study

What it is?

A case study is a type of research that focuses on **a small number of cases**, often even just one.

These cases are **chosen for a specific reason**, such as being unusual, important, or useful for answering a research question.

All cases usually show **the same outcome**, and the researcher studies them in depth.

Case studies **do not use experimental manipulation**; instead, they observe and analyze what has already happened.

The main goal is to **understand a specific case** as fully as possible.

Example(s)

- Studying one company to understand how it adapted to a crisis.
- Examining a single school known for unusually high performance.
- Researching one patient case to understand a rare medical condition.
- Analyzing a historical event to explain why it happened the way it did.

Multivariate study

What it is?

A multivariate study is research that uses a **large number of cases**, usually selected randomly or in another representative way so that they reflect the population.

The **dependent variable changes** across these cases, which allows researchers to compare different situations or conditions.

The study either includes **experimental manipulation** or tries to mimic it, for example through natural experiments.

The main goal is to **identify and understand general patterns** that apply to the broader population.

Example(s)

- Surveying thousands of people to study how income, education, and age affect voting behaviour.
- Comparing test scores across many schools to see how class size influences performance.
- Studying medical data from many patients to understand which factors increase disease risk.

Analytical narrative

What it is?

An analytical narrative is a type of case study that uses a **story-like explanation** to show how key actors, their preferences, and their decisions led to a particular outcome.

It focuses on identifying the **important actors**, their **main decision points**, and the logic behind their choices, often using ideas from game theory or rational choice.

The goal is to make the reasoning in the case clear and to generate explanations that can be evaluated.

Example(s)

- Telling the story of why a government chose a specific policy by describing the actors involved and the decisions they faced.
- Explaining why a company succeeded by outlining the sequence of strategic choices made by its leaders.
- Describing why a historical rebellion occurred by identifying the key groups and the decisions that shaped the event.

Qualitative comparative analysis

What it is?

Qualitative comparative analysis (QCA) is a method where several cases are compared to find out which conditions are **needed** for an outcome

to happen.

It helps researchers understand **when and why** a result occurs.

The findings can also be used later in larger, statistical studies.

Example(s)

- Comparing different successful strikes to see which conditions must always be present.
- Looking at several failing companies to find a factor that appears in every failure.
- Studying high-performing schools to see which condition is necessary for success.

Process tracing

What it is?

Process tracing is a type of case study that investigates **cause-and-effect inside one single case**, not many cases like Qualitative Comparative Analysis (QCA).

It tries to uncover the **exact sequence of steps**—the mechanism—that produced the outcome in that specific case.

Instead of looking for patterns across many cases or calculating average effects in a population, process tracing focuses on **how** and **why** the outcome happened in this one situation.

The goal is to identify the **causal mechanism**: the chain of events linking the cause to the final result.

Example(s)

- Studying a single political crisis to see which decisions, mistakes, or pressures directly caused the government to fall.

- Examining one company’s success by identifying each strategic move that led to its growth.
- Investigating a historical event, such as a revolution, by tracing the key moments and actions that built up to the final outcome.

L10. Research Ethics

Morality

Descriptive view of morality

What it is?

The descriptive view of morality says that morality is simply the rules and codes of conduct that people actually accept. This view is problematic because many accepted rules are not moral, and it leads to relativism where no action can be judged as truly morally wrong. Another way to define it as a code of conduct that is put forward by a society, group, or individual, that is accepted as a guide to behaviour

Example(s):

- Driving on the right side of the road is a shared rule, but breaking it is not immoral, only a mistake.
- Etiquette rules, like not speaking with your mouth full, are not moral rules.
- If morality were only based on accepted codes, even clearly unethical actions could not be condemned if a group accepted them.

Normative View

What it is?

The **normative view of morality** says that morality is not based on what people actually do or believe. Instead, it is based on what **all rational people would agree is right or wrong if they were thinking clearly and fairly**. These are ideal conditions where people do not know their own interests, power, or social position. This makes morality objective rather than relative to cultures or groups. It also allows us to criticise laws, traditions, or scientific practices even if many people accept them.

Example(s):

- **Harming others:** Rational people, thinking impartially, would agree that harming innocent people is wrong, no matter where or when it happens.
- **Scientific experiments:** Experiments that seriously harm participants are morally wrong even if scientists or governments approve of them.
- **Law vs morality:** A law can exist and be followed, but still be immoral if rational people would reject it under fair conditions.
- **Research ethics:** Rules like informed consent and avoiding unnecessary harm are moral rules because rational people would support them, not just because they are written in guidelines.

Ethics

What it is?

Ethics is a theory that explains **which rules of behaviour we should follow and why**. It gives clear, rational reasons for judging actions as right or wrong. Ethics is not about describing what people do, but about **justifying what people ought to do**.

Example(s):

- **Research ethics:** Explains why harming research participants is wrong and why informed consent is required.
- **Honesty:** Gives reasons why lying is wrong, even if it is common or useful.
- **Engineering practice:** Justifies why safety should matter more than saving money.

Moral vs Ethics

What it is?

The **moral vs ethical** distinction says that **morals** are about whether an action, intention, or decision is **good or bad**, while **ethics** is about **explaining and justifying why** it is good or bad. Morals describe the moral status of actions. Ethics gives the reasons and theories behind that judgment. In everyday language, this difference is often not kept, and the words are used as if they mean the same thing.

Example(s):

- **Morals:** Lying to a friend is morally wrong.
- **Ethics:** Ethics explains why lying is wrong, for example because it breaks trust or harms others.
- **Everyday language:** The same action might be called *immoral* or *unethical* with no real difference in meaning.

Three Frameworks of Normative Ethics

Deontology

What it is?

Deontology is an ethical view that says an action is right or allowed if it **follows certain duties or rules**. These duties come from roles we have or from moral obligations, such as **respecting the rights of other people**. The focus is on what we must do, not on the outcomes.

Example(s):

- **Respecting rights:** It is wrong to harm someone because we have a duty to respect their rights.
- **Telling the truth:** Lying is wrong because honesty is a duty, even if lying could lead to good results.
- **Professional roles:** A doctor must respect patient autonomy because it is part of their duty.

Consequentialism

What it is?

Consequentialism is an ethical view that says an action is right or wrong **only because of its consequences**. The action itself does not

matter, only the results it leads to. A common form is **utilitarianism**, which aims to produce the most happiness or overall benefit.

Example(s):

- **Saving many people:** An action is right if it saves more lives than the alternatives.
- **Policy decisions:** A law is good if it increases overall well-being, even if some individuals are harmed.
- **Lying:** Lying can be acceptable if it leads to better outcomes than telling the truth.

Virtue ethics

What it is?

Virtue ethics is an ethical view that focuses on **character rather than rules or outcomes**. An action is right if it shows good character traits, called virtues, such as honesty, courage, humility, and respect. The goal is to become a good person and to develop human abilities in a balanced way.

Example(s):

- **Honesty:** Telling the truth because honesty is a virtue of a good person.
- **Courage:** Acting bravely in a difficult situation, even when it is risky.
- **Professional life:** A scientist acting carefully and responsibly because integrity is part of good character.

Categorical Imperative

What it is?

The **categorical imperative** is a rule in **Kant's ethics**. It says that an action is right only if it follows a **principle that everyone could reasonably accept and follow**. The action must be acceptable as a general rule, not just useful in one case.

Example(s):

- **Lying:** Lying is wrong because if everyone lied, trust would break down, so the principle cannot be defended for everyone.

- **Keeping promises:** Keeping promises is right because it can be accepted as a rule for all people.
- **Respecting people:** Using others only as tools is wrong because no one could accept that as a general principle.

Informed Consent

What it is?

Informed consent means that a person agrees to take part in an activity, such as research, **in a proper and ethical way**. This requires three things: the person is given enough information, joins by free choice, and is able to understand and decide.

Example(s):

- **Informed:** A research participant is told the purpose of the study, what will happen, and possible risks.
- **Voluntary:** The participant joins without pressure, threats, or rewards that force the decision.
- **Decision-capacitated:** The participant is mentally able to understand the information and make a clear decision.

Scientific Misconduct

Fabrication

What it is?

Fabrication is a form of scientific misconduct where a researcher **intentionally invents data or results**. The information is not based on real experiments or observations and is created to **mislead others**.

Example(s):

- **Fake data:** Reporting measurements that were never actually collected.
- **Invented results:** Claiming that an experiment produced certain outcomes when the experiment was never done.
- **Misleading publications:** Publishing graphs or tables based on made-up numbers.

Falsification

What it is?

Falsification is a form of scientific misconduct where a researcher **intentionally changes, manipulates, or removes data or results** without scientific justification. The goal is to **mislead the reader** and make the results look better or more convincing than they really are.

Example(s):

- **Changing data:** Altering measurements so they better fit a hypothesis.
- **Removing results:** Leaving out data points that contradict the expected outcome.
- **Manipulating images:** Editing images or graphs to exaggerate effects or hide problems.

Plagiarism

What it is?

Plagiarism is a form of academic misconduct where someone **uses another person's ideas, methods, results, or words as if they were their own**, without proper acknowledgement. It misleads others about who did the original work.

Example(s):

- **Copying text:** Using sentences from a paper or book without citing the source.
- **Stealing ideas:** Presenting another researcher's idea or argument without credit.

- **Uncredited results:** Using data or findings from someone else's study without mentioning them.

Whistleblowing

What it is?

Whistleblowing means **reporting illegal or unethical actions** that happen inside an organisation. It is usually done by someone who works in or is closely connected to the organisation, with the aim of stopping harm or wrongdoing.

Example(s):

- **Research misconduct:** A scientist reports fabricated or falsified data in their lab.
- **Safety issues:** An engineer reports that a company is hiding serious safety risks.
- **Corruption:** An employee reports fraud or misuse of public funds.

Authorship and some variants

Author

What it is?

An **author** is someone who has made a **major and responsible contribution** to a scientific or academic work. To be an author, the person must contribute to the idea, help write the work, approve the final version, and take responsibility for the whole content.

Example(s):

- **Research design:** A researcher helps plan the study and its main ideas.

- **Writing:** A contributor helps draft or revise the article in a meaningful way.
- **Final approval:** The person reads and approves the final version before submission.
- **Responsibility:** The author accepts responsibility for the accuracy and integrity of the entire work.

Gift Authorship

What it is?

Gift authorship means listing someone as an author **even though they did not make a real contribution** to the work. This is often done for social reasons, such as politeness, hierarchy, or gaining favour, and it is considered unethical.

Example(s):

- **Senior pressure:** Adding a supervisor as an author even though they did not help with the research or writing.
- **Favour exchange:** Listing a colleague as an author to return a favour.
- **Status boosting:** Including a well-known name to make the paper seem more important.

Ghost authorship

What it is?

A **ghost author** is a person who **made a substantial contribution to a work but is not listed as an author**. This hides who actually did the work and is considered unethical.

Example(s):

- **Uncredited writer:** A researcher writes large parts of a paper but is not named as an author.
- **Industry influence:** A company employee helps write a scientific article, but only academic researchers are listed.

- **Student work:** A student does most of the research and writing, but only a supervisor is named as author.

Frivolous Research

What it is?

Frivolous research is research done **mainly for entertainment or curiosity**, without a serious scientific, practical, or social purpose. As far as one can tell, the goal is not to gain useful knowledge or solve an important problem.

Example(s):

- **Playful experiments:** Studying a topic only because it is amusing, with no clear research value.
- **Pointless measurements:** Collecting data that does not help answer any meaningful question.
- **Novelty projects:** Doing research mainly to be funny or unusual, not to contribute to science or society.

Effective altruism

What it is?

Effective altruism is a way of thinking and a movement that aims to **help others as much as possible**. It focuses on using **evidence and careful reasoning** to choose actions that do the most good with the resources available.

Example(s):

- **Charity choice:** Donating money to causes that save or improve the most lives per krona or dollar.
- **Career choice:** Choosing a job where you can have the biggest positive impact on others.
- **Global health:** Supporting programs like vaccinations because they are highly effective at reducing suffering.

L.11. Anticipating Risk In Science and Engineering

Definitions of Risk

Risk as an Unwanted Event

What it is?

Risk, seen as an unwanted event, means that something bad can happen, but it does not have to happen. The focus is on the bad event itself and the uncertainty around it. A situation involves risk when there is a chance that a harmful or negative event will occur. This way of understanding risk is common in everyday life and in simple risk analysis.

Example(s):

- Damage to the lungs from breathing in smoke is a big risk in fires.
- A chemical leak from a factory is a risk for people living nearby.
- Losing important files is a risk if computers are not backed up.
- Buildings collapsing is a risk during earthquakes.

Risk as a Cause of an Event

What it is?

Risk, understood as a cause of an unwanted event, refers to something that increases the chance that a bad event will happen. The focus is not on the bad event itself, but on the factor that can lead to it. In this view, a risk is a condition, action, or exposure that can cause harm. If the cause is present, the unwanted event becomes more likely.

Example(s):

- Smoking is a health risk because it can cause diseases like cancer.

- Driving too fast is a risk because it can lead to accidents.
- Using weak passwords is a risk because it can cause data theft.
- Working without safety equipment is a risk because it can lead to injuries.

Risk as Spread of Outcomes

What it is?

Risk, understood as the spread of outcomes, refers to how much possible results can vary. The focus is not only on bad outcomes, but on the fact that results can be very different from each other. In this view, a situation is more risky when the possible outcomes range from very good to very bad, and when it is hard to know which outcome will occur. This way of thinking about risk is common in finance and investments.

Example(s):

- A stock investment is risky because it can lead to high gains or large losses.
- Starting a new business is risky because it might succeed greatly or fail completely.
- Investing in cryptocurrency is risky due to large price changes.
- Choosing a variable interest loan is risky because payments can increase or decrease a lot.

Quantitative Notions of Risk

What it is?

Risk, in a quantitative sense, is defined using numbers. It can mean how likely an unwanted event is to happen, or how serious the consequences would be if it happens. Sometimes risk is described only by probability, and sometimes only by severity. In other cases, both are considered together. This way of defining risk is common in engineering, science, and risk analysis because it allows comparison and calculation.

Example(s):

- There is a 5% risk that a machine will fail during operation.
- The risk of flooding is high because the damage would be very severe.
- A medical treatment has a low risk of side effects, meaning the chance is small.
- An industrial accident is considered a high risk because it could cause many injuries.

Qualitative Notions of Risk

What it is?

Qualitative risk means describing risk using words instead of numbers. Risks are judged as high, medium, or low based on experience, expert judgement, or general understanding. This approach is useful when there is not enough data to calculate probabilities or when risks are hard to measure exactly.

Example(s):

- Working at great heights is considered a high risk job.
- Walking alone at night in an unsafe area is a high risk situation.
- A small software bug is seen as a low risk problem.
- Using new and untested technology is often described as a medium or high risk.

Decision Making Under Different Conditions

Decision-Making Under Certainty

What it is?

Decision-making under certainty means that there is only one possible outcome for an action. If you choose a certain action, you know exactly what will happen. There is no risk or uncertainty because the result is fixed and known in advance.

Example(s):

- If you press a light switch, the light turns on.
- Paying a fixed price for a bus ticket means you know the exact cost.
- Setting an alarm for 7:00 means it will ring at that time.
- Buying a product with a fixed price gives a known result: you pay that price and get the item.

Decision-Making Under Risk

What it is?

Decision-making under risk means that an action can lead to different outcomes, but the chances of each outcome are known. You do not know which outcome will happen, but you know how likely each one is. This allows decisions to be made using probabilities.

Example(s):

- Rolling a dice, where each number has a known chance of appearing.
- Playing a lottery, where the chance of winning is known.
- An insurance company knowing the probability of car accidents.
- Choosing a medical treatment when the success rate is known.

Decision-Making Under Ignorance

What it is?

Decision-making under ignorance means that an action can lead to several possible outcomes, but the chances of these outcomes are not known. You know what might happen, but you do not know how likely each outcome is. Because of this, it is harder to compare options using numbers.

Example(s):

- Launching a completely new product in an unknown market.
- Exploring a new technology that has never been tested before.
- Investing in a new company with no past data.

- Making decisions during a natural disaster when little information is available.

Decision-Making Under Deep Uncertainty

What it is?

Decision-making under deep uncertainty means that we do not know all possible outcomes, and we also do not know their probabilities. Some outcomes may be completely unexpected. This happens in complex or new situations where knowledge is very limited. In these cases, normal risk calculations do not work well.

Example(s):

- Making long-term climate policy decisions.
- Developing new technologies with unknown future effects.
- Planning for pandemics before they occur.
- Managing nuclear waste over very long time periods.

Expected Value Models of Risk

What it is?

Expected value models describe risk using numbers for both parts of an outcome: how likely it is to happen and how serious it would be if it happens. Each possible outcome has a probability and a level of damage or benefit. These models combine probability and severity to compare different choices and decide which option has the lowest or highest expected result.

Example(s):

- An insurance company calculates expected losses by combining accident probability with repair costs.

- A company compares projects by looking at expected profit and expected loss.
- A government estimates expected damage from floods using chance and damage size.
- A casino calculates expected winnings and losses for different games.

Decision Strategies

Principle of Insufficient Reason

What it is?

The principle of insufficient reason says that if we do not know how likely different outcomes are, we should treat them as equally likely. When there is no good reason to think one outcome is more likely than another, we assign the same probability to each. This is a simple way to make decisions when probability information is missing.

Example(s):

- If a box could contain one of four prizes and there is no information about which one it is, each prize is given a 25% chance.
- When choosing between two unknown options with no data, each is treated as equally likely.
- If a coin is found and nothing is known about it, one may assume heads and tails are equally likely.

Minimax

What it is?

Minimax is a decision rule used when outcomes are uncertain. It means choosing the action where the worst possible result is better than the worst results of the other options. The focus is on avoiding the most

serious loss, even if this means giving up the chance of a very good outcome.

Example(s):

- Choosing a job with a stable but lower salary instead of a risky job that could fail badly.
- Building a strong dam to reduce the worst possible flood damage, even if floods are unlikely.
- Saving money in a bank account instead of investing in a risky stock market.

Safety Factors

What it is?

Safety factors are used in engineering to make systems stronger than they need to be under normal conditions. Engineers add extra strength or capacity so the system can handle unexpected loads, stress, or errors. This helps reduce the chance of failure when something goes wrong or when conditions are worse than expected.

Example(s):

- A bridge is built to hold much more weight than normal traffic requires.
- A building is designed to withstand stronger winds than usually expected.
- An elevator cable is made much stronger than the maximum load it will carry.
- Aircraft parts are designed to handle stresses beyond normal use.

Precautionary Principle

What it is?

The precautionary principle says that if an action or policy might cause serious harm, we should be careful, even if there is no clear scientific agreement yet. When there is uncertainty, the responsibility is on those who want to take the action to show that it is safe. The idea is to prevent harm before it happens, rather than reacting after damage is done.

Example(s):

- Restricting a new chemical until it is shown to be safe for humans and the environment.
- Limiting the use of new technology when its long-term effects are unknown.
- Stopping the release of a new drug until enough testing proves it is not harmful.
- Regulating genetically modified organisms until their risks are better understood.

Subjective Evaluations of Risk

What it is?

Subjective evaluation of risk means that people judge risk based on their own feelings, values, and preferences, not only on numbers. People are often risk-averse, meaning they prefer safer options even if a risky option has a higher expected value. The St. Petersburg paradox shows that people do not always follow expected monetary value when making decisions. Daniel Bernoulli explained this by using *utility* instead of money, meaning that extra money matters less when you already have a lot. Risk judgments are also affected by whether a risk feels voluntary, familiar, or under personal control.

Example(s):

- Many people avoid gambling even if the expected gain is positive.
- People feel less worried about driving a car than flying, even though flying is statistically safer.

- Someone may reject a risky investment because they dislike uncertainty, even if the numbers look good.
- A person may accept more risk in activities they choose freely, like extreme sports, than in risks imposed on them, like pollution.

Utilitarian Approaches to Risk (Also known as Actualist Approach)

What it is?

In utilitarianism, actions are judged by their outcomes and by how much good or harm they produce. When outcomes are uncertain, it is not obvious how actions should be evaluated. Two main approaches are often discussed: the **actualist approach** and the **expected utility approach**.

According to the **actualist approach**, an action is judged only by the outcome that actually happens. Possible outcomes that did not occur do not matter. If the final result is good, the action is considered right, even if there was a high risk of a bad outcome.

According to the **expected utility approach**, an action is judged by its expected utility. This means that all possible outcomes are considered, together with how likely they are and how good or bad they would be. An action is morally right if it has the highest expected utility, even if the actual outcome later turns out to be bad.

The actualist approach is controversial because it makes moral judgment depend on luck. It ignores the risks that a person knowingly accepted. The expected utility approach avoids this problem by focusing on the decision itself and the information available at the time, rather than on the final outcome alone.

Example(s):

- You decide not to reinforce a bridge even though there is a 50% chance it will collapse. You send a very heavy transport over it, and luckily the bridge does not collapse. According to the actualist approach, your decision was right because the outcome was good. According to the

expected utility approach, the decision was wrong because you accepted a very high risk of serious harm.

- A doctor chooses a risky treatment with a low chance of success. If the patient survives, the actualist approach says the decision was right. The expected utility approach judges the decision based on the success rate and possible harm, not on the lucky outcome.
- A company ignores safety warnings and no accident happens. The actualist approach sees this as acceptable, while the expected utility approach may judge it as wrong because the expected harm was high.

Deontological Approach to Risk

What it is?

The deontological approach says that actions should be judged by rules and rights, not only by their outcomes. People have rights that must not be violated, even if violating them could lead to good results. When applied to risk, this raises the question of whether exposing someone to a risk of harm already violates their rights, even if the harm never actually happens.

In this view, even a small chance of serious harm can be morally wrong if it violates someone's rights. However, how rights are understood can depend on the situation. Important factors include whether the risk is taken voluntarily, whether the exposure to risk is intentional or unintentional, and whether the person exposed to the risk also benefits from it.

Example(s):

- Exposing people to dangerous pollution without their consent may violate their rights, even if no one becomes ill.
- A person who chooses to take part in a risky sport accepts the risk voluntarily, which may make it less of a rights violation.
- Forcing workers to take serious risks without proper protection can be wrong, even if no accidents occur.

- Medical experiments without informed consent are considered wrong, even if they lead to useful results.
